Data Warehouse and Implementation

Andrew Fox & Mesgana Araia

OncoTrace

A business model for tracking cancer risk

Project Definition

Cancer is one of the most deadly and prominent non-communicable diseases world wide. It is the leading cause of death globally with nearly 10 million deaths in 2020, with breast, lung, colon, prostate, skin, and stomach cancers causing the most diagnoses. In particular, lung and breast cancers were amongst the most deadly, accounting for a combined death toll of nearly 2.5 million. By definition, cancer is caused by the transformation of normal cells into tumor cells, eventually resulting in a malignant tumor. The transformation of cells from normal to cancerous is a consequence of a person's genetic makeup interacting with their environment.

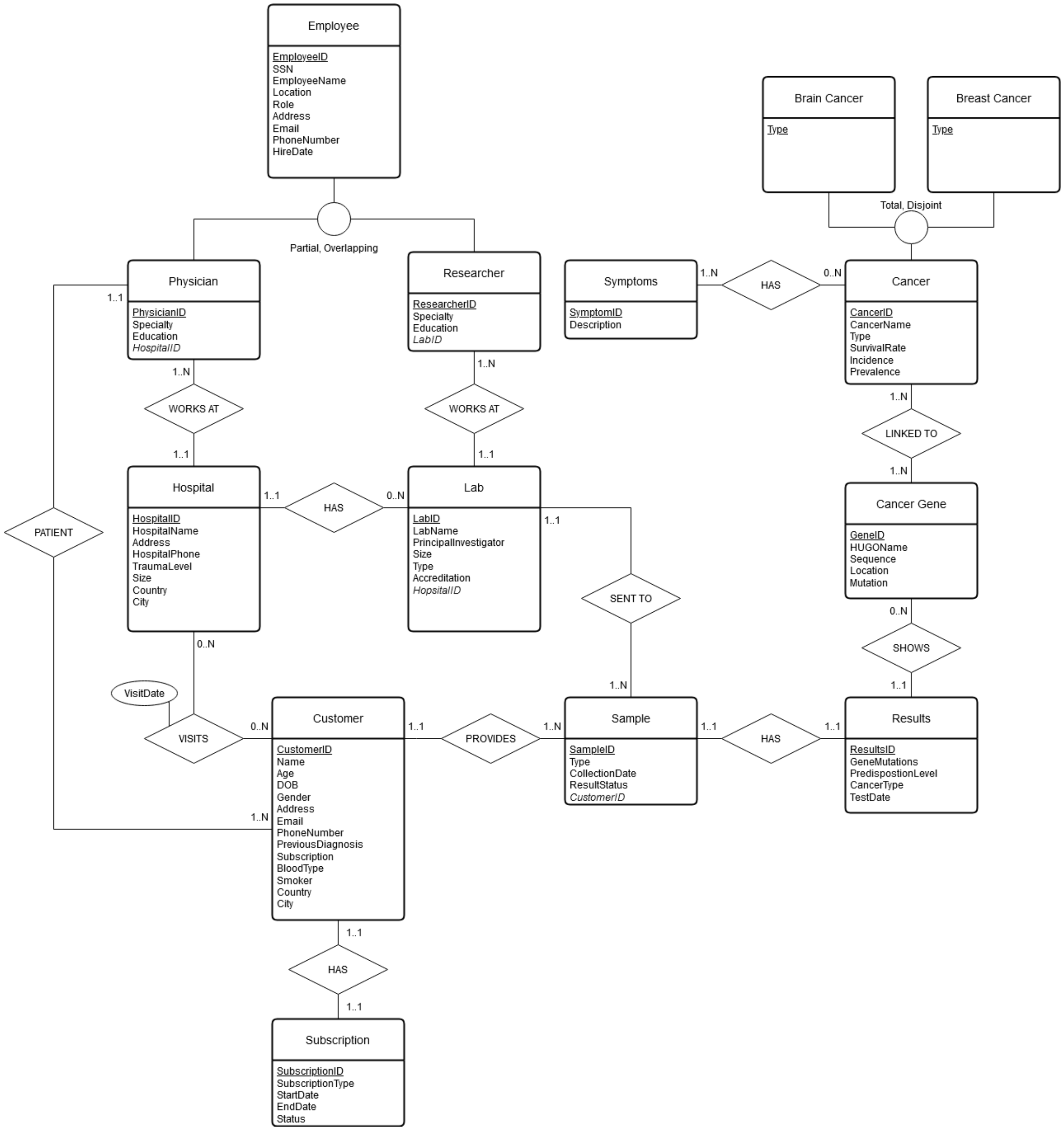
While cancer cannot be caused by a single factor, the genetic makeup of a person is crucial to the development of cancerous cells. Breast, ovarian (and other cancers related to male and female reproductive systems), leukemia, colorectal, brain and skin cancer are just some of the types of cancer that are highly linked to genetics. Thus, understanding one's genetic predisposition and altering external risk factors accordingly can significantly aid in reducing the potential for cancer development.

To facilitate this assessment, we developed a business model entitled OncoTrace, which collects customer samples and tests them for oncogenes or any other kinds of cancer-related genes (i.e. inactive tumor suppressing genes). OncoTrace provides customers with the knowledge of any cancerous risk and predispositions in a timely and cost effective manner. An understanding of genetic cancer risk can be crucial for customers in life and family planning.

OncoTrace allows customers to send in a sample and receive results, facilitating a quicker and cheaper alternative to traditional hospital or lab visits. The business would also provide recurring plans for customers to regularly send in samples, removing the hassle of repeated in-person visits and ideally expediting the process. This feature of OncoTrace gives customers with a family history of cancer peace of mind knowing that their predispositions are being tracked. In addition, OncoTrace can notice changes in the genetic makeup as they occur. Customers are made aware of these changes and can act promptly by contacting medical professionals. Catching cancer in its early stages drastically improves the patient's prognosis. Offering these services improves customers' mental and physical state, as they no longer have to worry about potential cancer progression and can catch cancer in its early stages.

The business would require a database to store and process customer data, business-related data, and of course biological and cancer-related data. A relational database could be beneficial. Certain entity types would include customers, scientists, samples (which may be saliva samples, nose or cheek swabs, skin samples, etc.), and a plethora of entities regarding the data on cancer, such as cancer types, predispositions, etc.

Entity-Relation Diagram



Relational Model

From the ERD we define the relational model. Primary keys are underlined. *Foreign keys* are italicized. All primary keys and foreign keys are declared NOT NULL.

Cancer (CancerID, CancerName, Type, SurvivalRate, Incidence, Prevalence)

Brain Cancer (Type, *CancerID*)
- FK *CancerID* references PK CancerID in **Cancer**

Breast Cancer (Type, *CancerID*)
- FK *CancerID* references PK CancerID in **Cancer**

Symptoms (SymptomID, Description)

Cancer Symptoms (*SymptomID*, *CancerID*)
- FK *SymptomID* references SymptomID in **Symptoms**
- FK *CancerID* references PK CancerID in **Cancer**

Hospital (HospitalID, HospitalName, Address, HospitalPhone, TraumaLevel, Size, Country, City)

Lab (LabID, *HospitalID*, LabName, PrincipalInvestigator, Size, Type, Accreditation)
- FK *HospitalID* refers PK HospitalID in **Hospital**

Employee (EmployeeID, SSN, EmployeeName, Location, Role, Address, Email, PhoneNumber, HireDate)

Physician (*PhysicianID*, Specialty, Education)
- FK *PhysicianID* references PK EmployeeID in **Employee**

Researcher (*ResearcherID*, Specialty, Education)
- FK *ResearcherID* references PK EmployeeID in **Employee**

Customer (CustomerID, Name, Age, DOB, Gender, Address, Email, PhoneNumber, PreviousDiagnosis, Subscription, BloodType, Smoker, Country, City, *PhysicianID*)
- FK *PhysicianID* references PK *PhysicianID* in **Physician**

Visits (VisitID, *HospitalID*, *CustomerID*, VisitDate)
- FK *HospitalID* references PK HospitalID in **Hospital**
- FK *CustomerID* references PK CustomerID in **Customer**

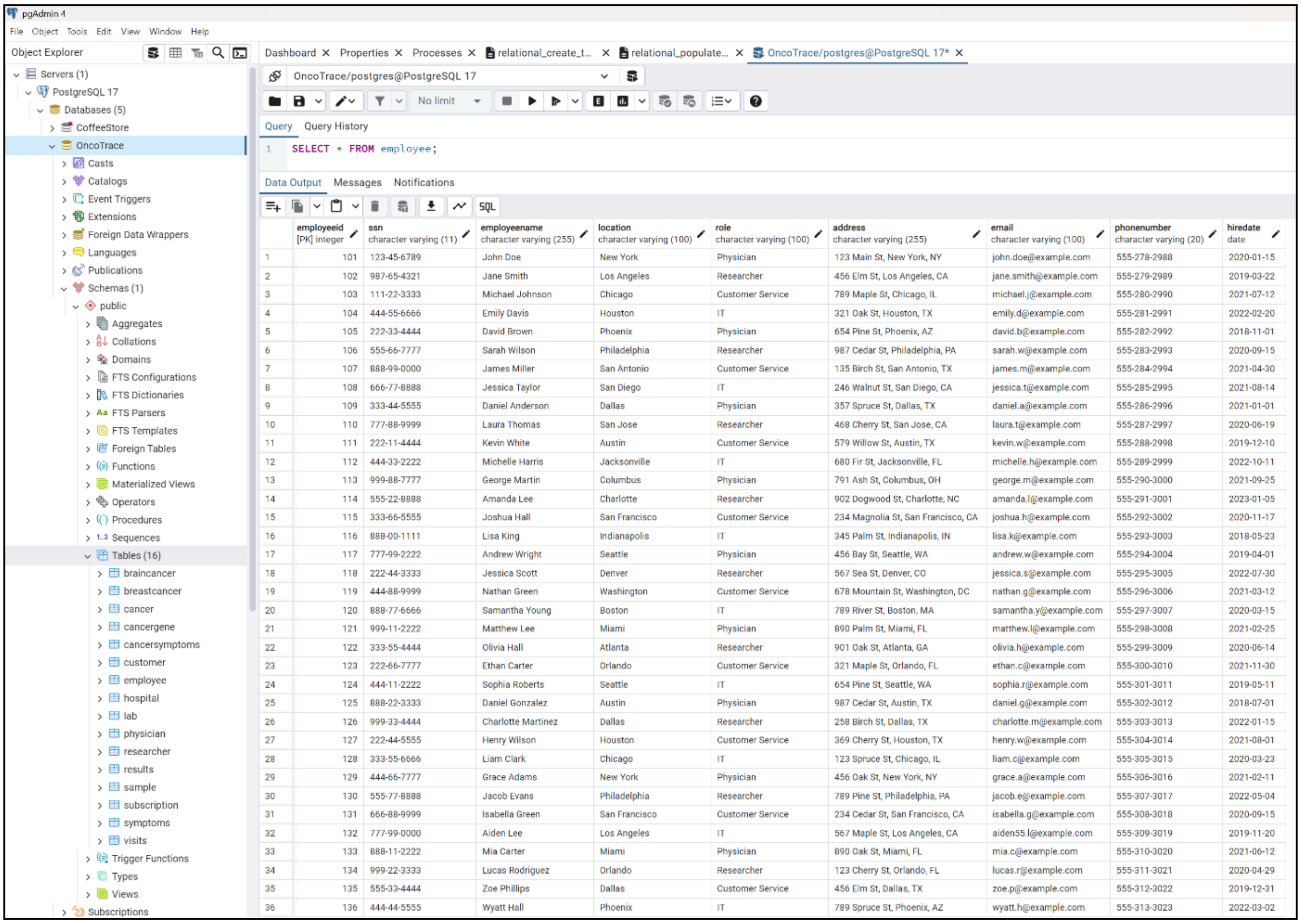
Subscription (SubscriptionID, SubscriptionType, StartDate, EndDate, Status, *CustomerID*)
- FK *CustomerID* references PK CustomerID in **Customer**

Sample (SampleID, Type, CollectionDate, ResultStatus, *CustomerID*, *LabID*)
- FK *CustomerID* references PK CustomerID in **Customer**
- FK *LabID* references PK LabID in **Lab**

Results (ResultID, GeneMutations, PredispositionLevel, CancerType, TestDate, *SampleID*)
- FK *SampleID* references PK SampleID in **Sample**

Cancer Gene (GeneID, HUGOName, Sequence, Location, Mutation, *CancerID*)
- FK *CancerID* references PK CancerID in **Cancer**

Implementation in PostgreSQL



	employeeid [PK] integer	ssn character varying (11)	employeename character varying (255)	location character varying (100)	role character varying (100)	address character varying (255)	email character varying (100)	phonenumber character varying (20)	hiredate date
1	101	123-45-6789	John Doe	New York	Physician	123 Main St, New York, NY	john.doe@example.com	555-278-2988	2020-01-15
2	102	987-65-4321	Jane Smith	Los Angeles	Researcher	456 Elm St, Los Angeles, CA	jane.smith@example.com	555-279-2989	2019-03-22
3	103	111-22-3333	Michael Johnson	Chicago	Customer Service	789 Maple St, Chicago, IL	michael.j@example.com	555-280-2990	2021-07-12
4	104	444-55-6666	Emily Davis	Houston	IT	321 Oak St, Houston, TX	emily.d@example.com	555-281-2991	2022-02-20
5	105	222-33-4444	David Brown	Phoenix	Physician	654 Pine St, Phoenix, AZ	david.b@example.com	555-282-2992	2018-11-01
6	106	555-66-7777	Sarah Wilson	Philadelphia	Researcher	987 Cedar St, Philadelphia, PA	sarah.w@example.com	555-283-2993	2020-09-15
7	107	888-99-0000	James Miller	San Antonio	Customer Service	135 Birch St, San Antonio, TX	james.m@example.com	555-284-2994	2021-04-30
8	108	666-77-8888	Jessica Taylor	San Diego	IT	246 Walnut St, San Diego, CA	jessica.t@example.com	555-285-2995	2021-08-14
9	109	333-44-5555	Daniel Anderson	Dallas	Physician	357 Spruce St, Dallas, TX	daniel.a@example.com	555-286-2996	2021-01-01
10	110	777-88-9999	Laura Thomas	San Jose	Researcher	468 Cherry St, San Jose, CA	laura.t@example.com	555-287-2997	2020-06-19
11	111	222-11-4444	Kevin White	Austin	Customer Service	579 Willow St, Austin, TX	kevin.w@example.com	555-288-2998	2019-12-10
12	112	444-33-2222	Michelle Harris	Jacksonville	IT	680 Fir St, Jacksonville, FL	michelle.h@example.com	555-289-2999	2022-10-11
13	113	999-88-7777	George Martin	Columbus	Physician	791 Ash St, Columbus, OH	george.m@example.com	555-290-3000	2021-09-25
14	114	555-22-8888	Amanda Lee	Charlotte	Researcher	902 Dogwood St, Charlotte, NC	amanda.l@example.com	555-291-3001	2023-01-05
15	115	333-66-5555	Joshua Hall	San Francisco	Customer Service	234 Magnolia St, San Francisco, CA	joshua.h@example.com	555-292-3002	2020-11-17
16	116	888-00-1111	Lisa King	Indianapolis	IT	345 Palm St, Indianapolis, IN	lisa.k@example.com	555-293-3003	2018-05-23
17	117	777-99-2222	Andrew Wright	Seattle	Physician	456 Bay St, Seattle, WA	andrew.w@example.com	555-294-3004	2019-04-01
18	118	222-44-3333	Jessica Scott	Denver	Researcher	567 Sea St, Denver, CO	jessica.s@example.com	555-295-3005	2022-07-30
19	119	444-88-9999	Nathan Green	Washington	Customer Service	678 Mountain St, Washington, DC	nathan.g@example.com	555-296-3006	2021-03-12
20	120	888-77-6666	Samantha Young	Boston	IT	789 River St, Boston, MA	samantha.y@example.com	555-297-3007	2020-03-15
21	121	999-11-2222	Matthew Lee	Miami	Physician	890 Palm St, Miami, FL	matthew.l@example.com	555-298-3008	2021-02-25
22	122	333-55-4444	Olivia Hall	Atlanta	Researcher	901 Oak St, Atlanta, GA	olivia.h@example.com	555-299-3009	2020-06-14
23	123	222-66-7777	Ethan Carter	Orlando	Customer Service	321 Maple St, Orlando, FL	ethan.c@example.com	555-300-3010	2021-11-30
24	124	444-11-2222	Sophia Roberts	Seattle	IT	654 Pine St, Seattle, WA	sophia.r@example.com	555-301-3011	2019-05-11
25	125	888-22-3333	Daniel Gonzalez	Austin	Physician	987 Cedar St, Austin, TX	daniel.g@example.com	555-302-3012	2018-07-01
26	126	999-33-4444	Charlotte Martinez	Dallas	Researcher	258 Birch St, Dallas, TX	charlotte.m@example.com	555-303-3013	2022-01-15
27	127	222-44-5555	Henry Wilson	Houston	Customer Service	369 Cherry St, Houston, TX	henry.w@example.com	555-304-3014	2021-08-01
28	128	333-55-6666	Liam Clark	Chicago	IT	123 Spruce St, Chicago, IL	liam.c@example.com	555-305-3015	2020-03-23
29	129	444-66-7777	Grace Adams	New York	Physician	456 Oak St, New York, NY	grace.a@example.com	555-306-3016	2021-02-11
30	130	555-77-8888	Jacob Evans	Philadelphia	Researcher	789 Pine St, Philadelphia, PA	jacob.e@example.com	555-307-3017	2022-05-04
31	131	666-88-9999	Isabella Green	San Francisco	Customer Service	234 Cedar St, San Francisco, CA	isabella.g@example.com	555-308-3018	2020-09-15
32	132	777-99-0000	Aiden Lee	Los Angeles	IT	567 Maple St, Los Angeles, CA	aiden55.l@example.com	555-309-3019	2019-11-20
33	133	888-11-2222	Mia Carter	Miami	Physician	890 Oak St, Miami, FL	mia.c@example.com	555-310-3020	2021-06-12
34	134	999-22-3333	Lucas Rodriguez	Orlando	Researcher	123 Cherry St, Orlando, FL	lucas.r@example.com	555-311-3021	2020-04-29
35	135	555-33-4444	Zoe Phillips	Dallas	Customer Service	456 Elm St, Dallas, TX	zoe.p@example.com	555-312-3022	2019-12-31
36	136	444-44-5555	Wyatt Hall	Phoenix	IT	789 Spruce St, Phoenix, AZ	wyatt.h@example.com	555-313-3023	2022-03-02

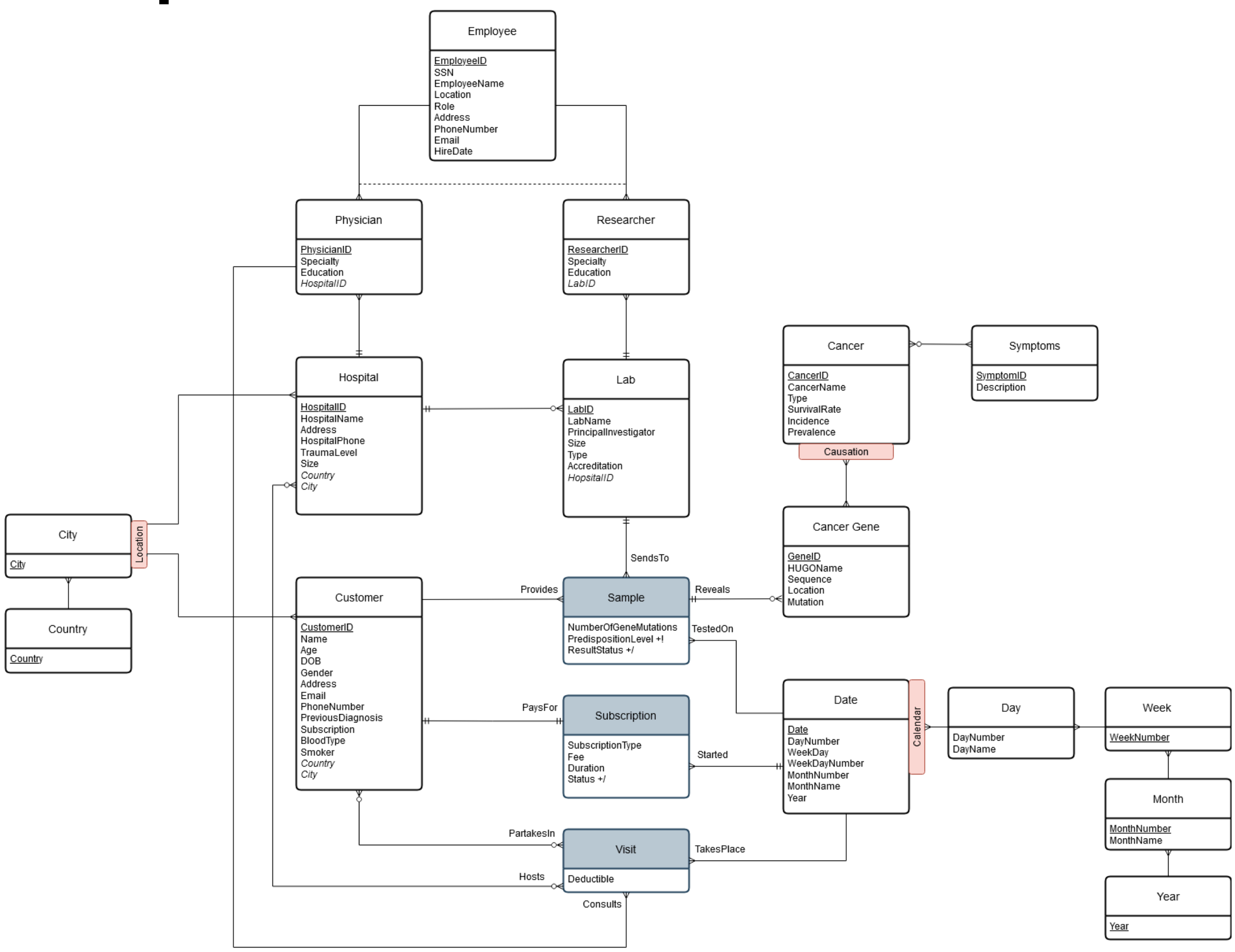
Data for the Employee, Physician, Researcher, Customer tables were generated using Mockaroo.com, with assistance from ChatGPT. Data for the Lab, Visits, Sample, Subscription, CancerGenes, CancerSymptoms, and Results tables were created using ChatGPT.

Data for the Hospital table was created manually. Hospitals were selected from a list here. Information on these hospitals such as their address or phone number were found on Google.

Data for the Cancer, BreastCancer, and BrainCancer tables were created manually. The brain and breast cancer types were selected from here and here, respectively. The Symptoms data was created manually and gathered from the cancer's respective MayoClinics page (1, 2, 3, 4, 5, 6).

While information on cancers and their symptoms (except survival rate, incidence, and prevalence) are scientifically accurate, any data on genes is entirely fabricated.

Conceptual Data Warehouse Model



Comments on MultiDim Model

The MultiDim Conceptual model for the data warehouse features 3 facts (Sample, Subscription, and Visit) and 16 dimensions (Employee, Physician, Researcher, Hospital, Lab, Customer, Symptoms, Cancer, Cancer Gene, Date, Day, Week, Month, Year, City, and Country).

- There is a Calendar hierarchy for the date consisting of 4 levels (Day, Week, Month, and Year).
- There is a Causation hierarchy for Cancer consisting of two levels (Cancer and Cancer Gene).
- There is a Location hierarchy consisting of two levels (Country and City).

The Sample fact has 3 measures. NumberOfGeneMutations is an additive measure for the number of gene mutations found in a sample. PredispositionLevel is a semi-additive measure for the predisposition to cancer that any found gene mutations has. ResultStatus is a non-additive binary measure for whether or not the sample tests positive for a cancer.

- A Lab can receive up to many Samples.
- A Customer provides up to many Samples over the course of their subscription.
- Many Samples may be tested on a Date.
- A Sample may reveal many CancerGenes.

The Subscription fact has 4 measures. SubscriptionType is a non-additive measure for the type of subscription (monthly, yearly). Fee is an additive measure for how much the subscription costs. Duration is an additive measure for how long the subscription has been or was active. Status is a non-additive measure for whether or not the subscription is still active or canceled.

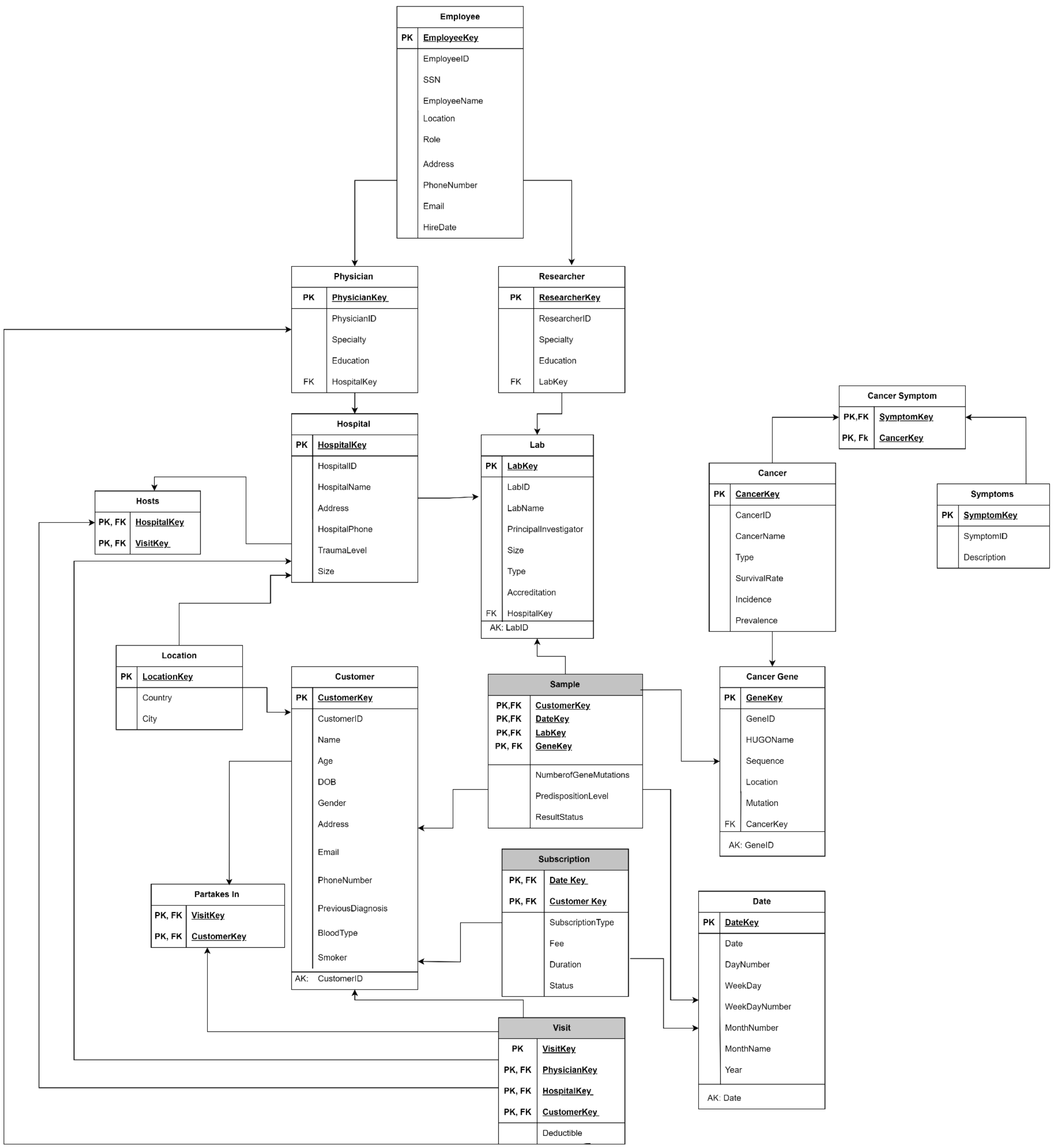
- A Subscription is paid for by a Customer.
- A Subscription is started on a Date.

The Visit fact has 1 measure. Deductible is an additive measure for how much the customer pays for that visit.

- A Visit is hosted by a Hospital.
- A Visit is consulted by a Physician.
- A Visit involves a Customer.
- A Visit takes place on a Date.

Logical Data Warehouse Model

Snowflake Model



Comments on Snowflake Model

Surrogate keys are generated for each dimension. They serve as the composite keys in the fact tables for levels participating in roles with those facts.

The Visit fact table has many-to-many relationships with the Customer and Hospital dimensions, which necessitates the use of bridge tables (Partakes In and Hosts). Visit's surrogate key is useful and helps in scalability in the future.

The Cancer Symptom table is another bridge table created to represent a many-to-many relationship between Cancer and Symptoms. It uses the primary keys of these dimensions as a composite primary key.

Implementation in PostgreSQL

By ETL in Talend

Screenshots were taken after jobs were run but also after Talend crashed, so the rows are not being shown. It wouldn't be ideal to run the jobs again because there are many surrogate key dependencies. And if run again records would flow in the found output, which does not exemplify the actual loading. For more insights into the Talend job please check the full Talend project

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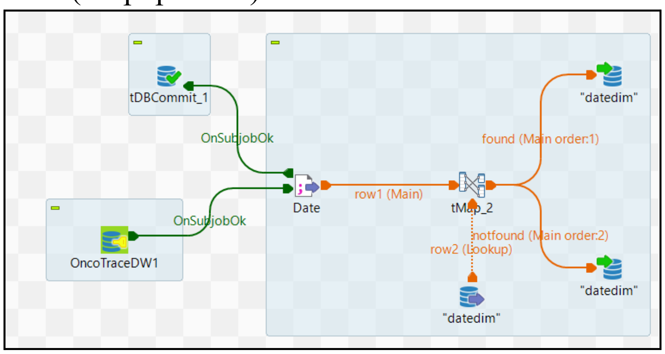
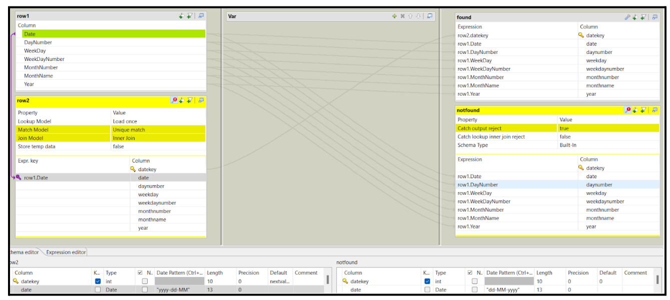
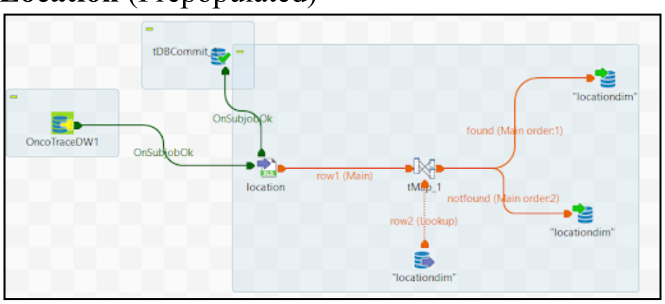
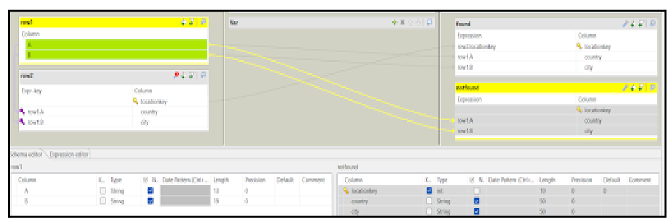
Date (Prepopulated)



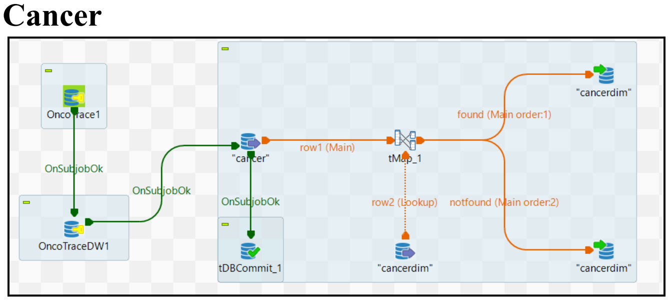


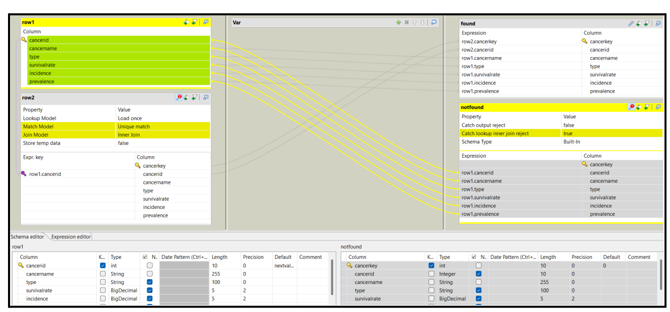
Location (Prepopulated)



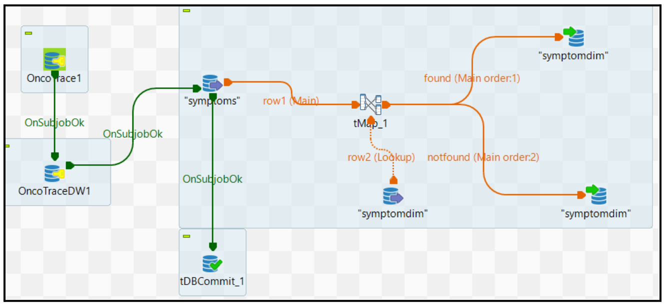


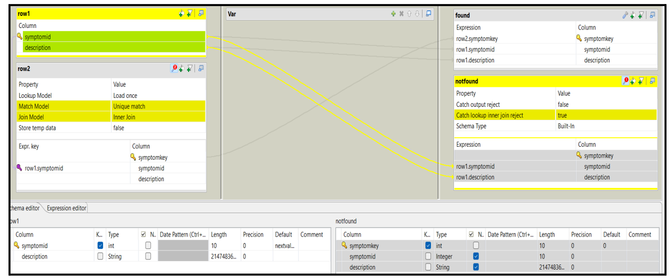
Cancer



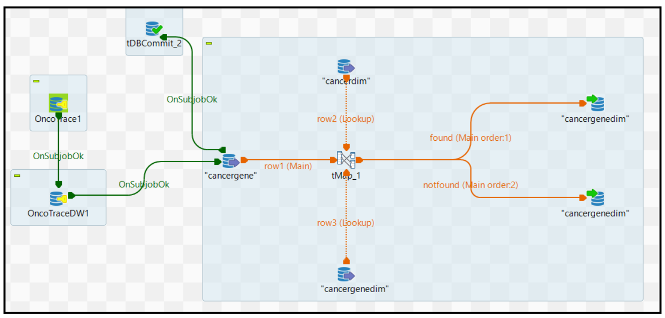


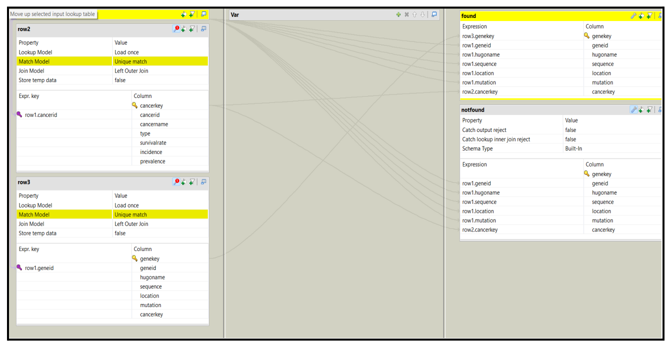
Symptom



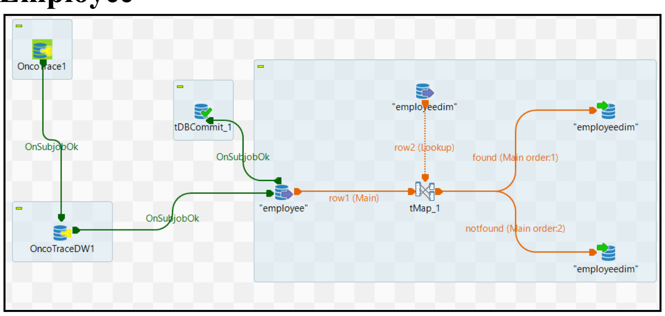


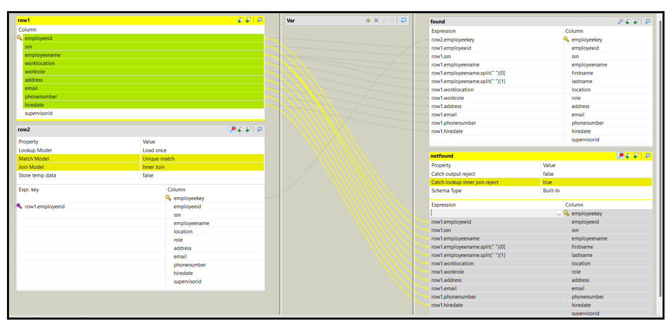
CancerGene



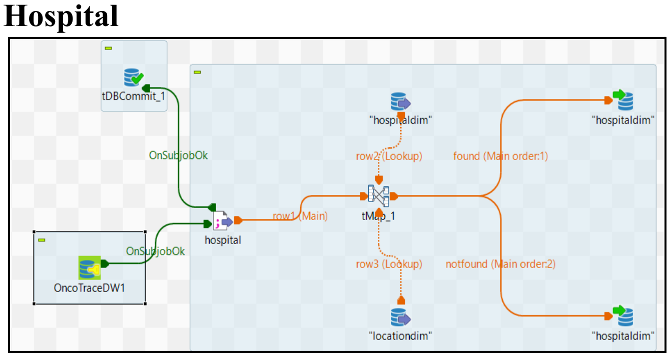


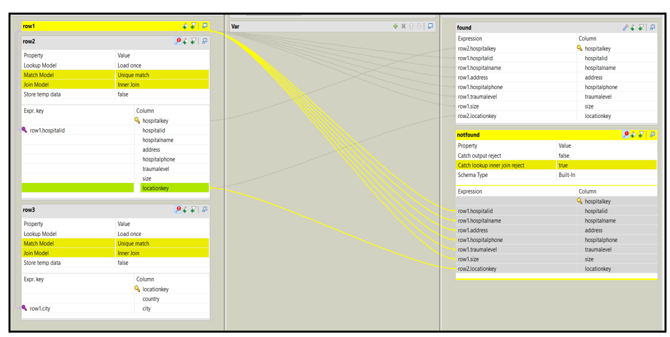
Employee



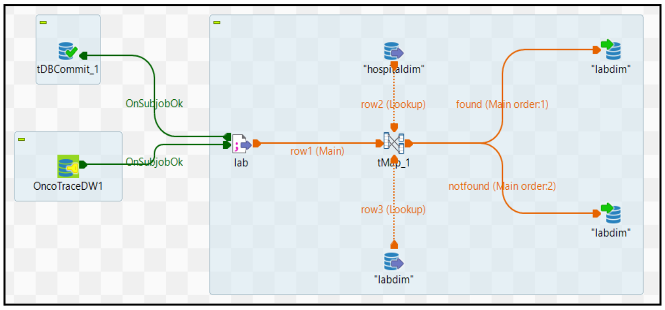


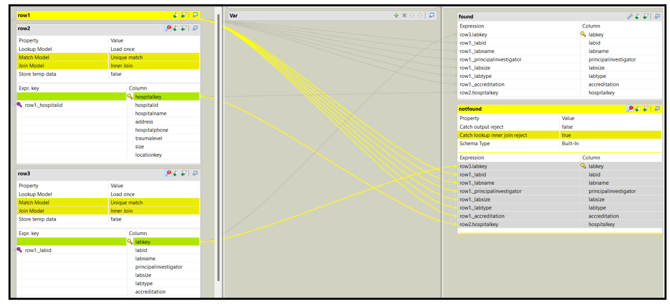
Hospital





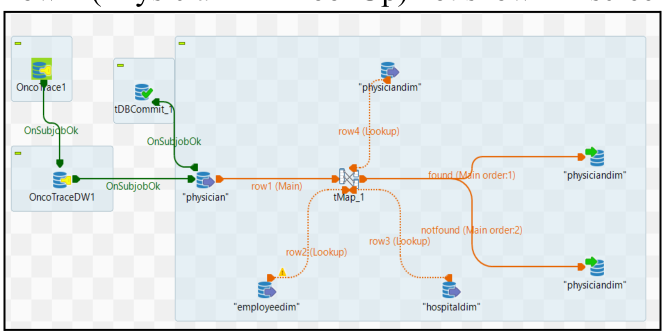
Lab

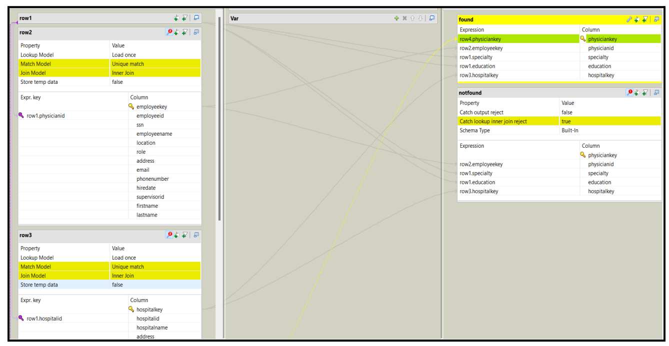




Physician

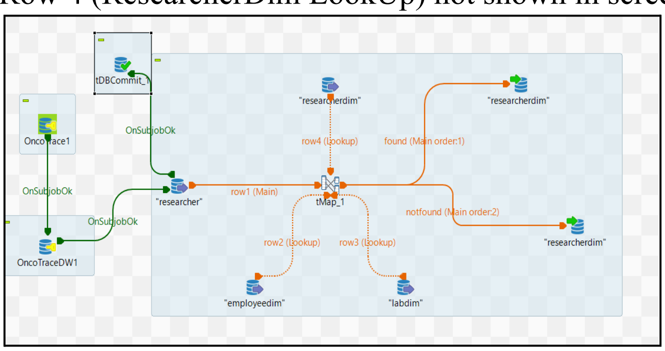
Row 4 (PhysicianDim LookUp) not shown in screenshot

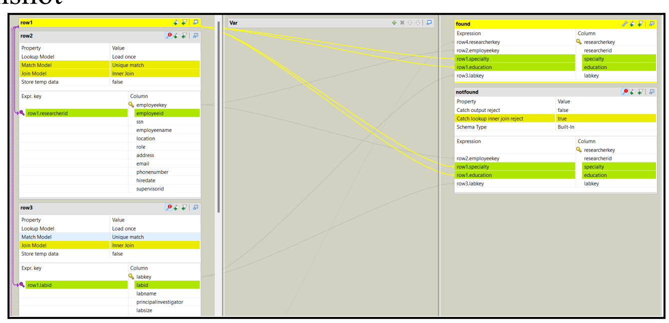




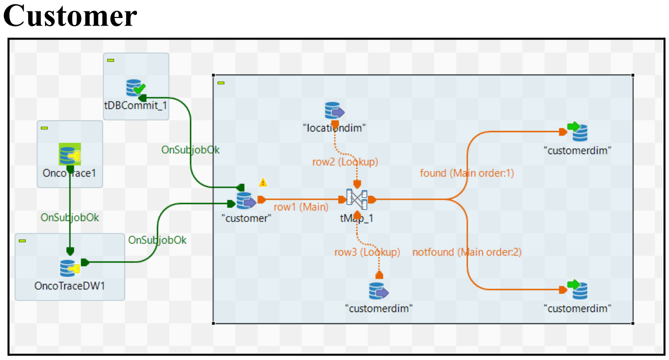
Researcher

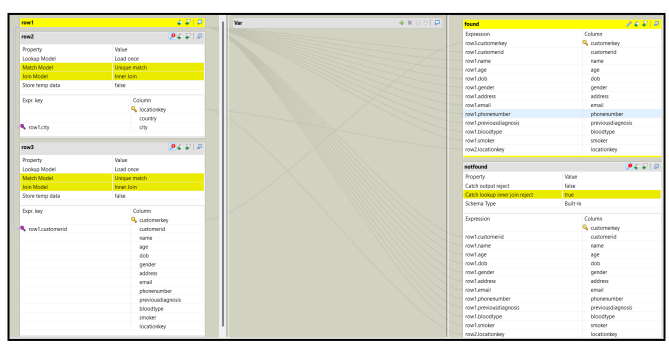
Row 4 (ResearcherDim LookUp) not shown in screenshot





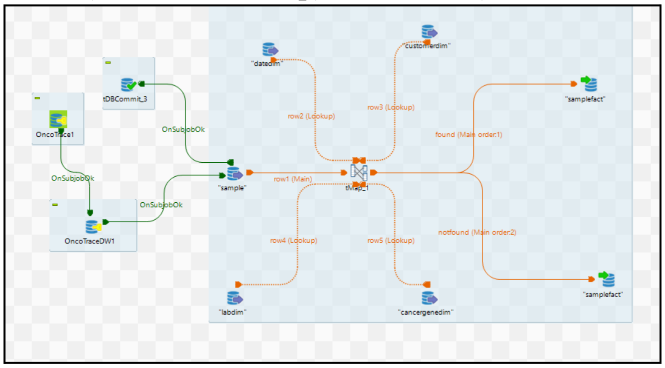
Customer

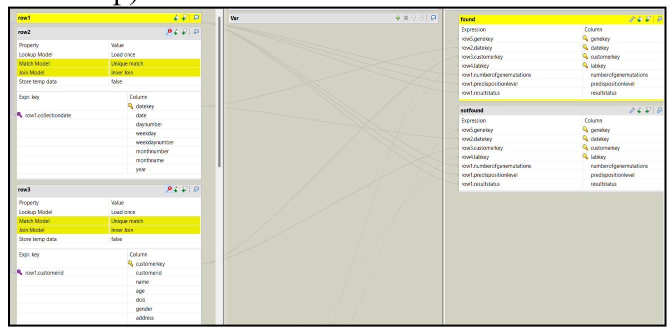




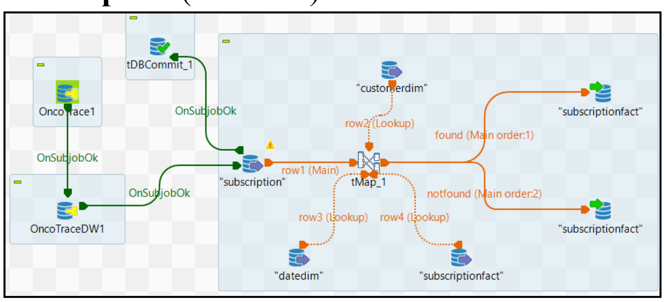
Sample (fact table)

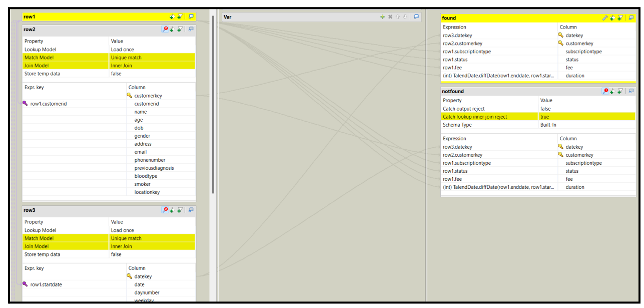
Row 4(LabDim LookUp) and Row 5(CancerGeneDim LookUp) not shown





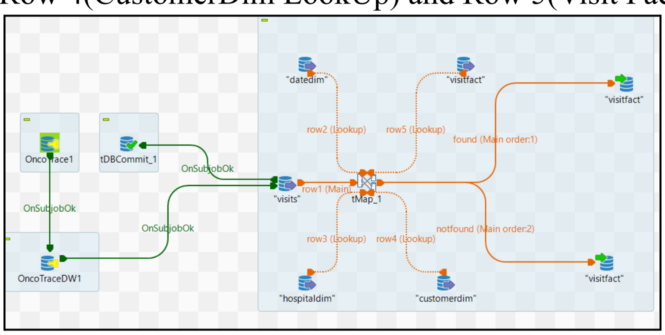
Subscription (fact table)

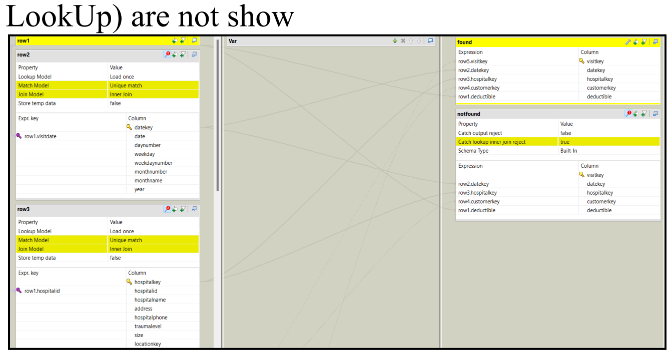




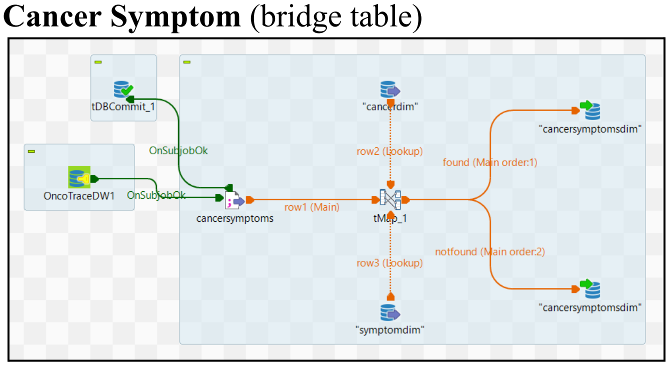
Visit (fact table)

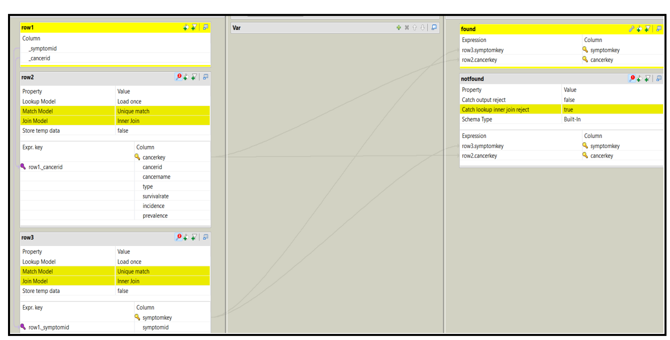
Row 4(CustomerDim LookUp) and Row 5(Visit Fact LookUp) are not show



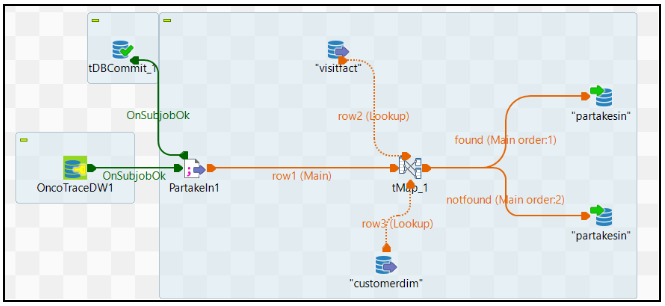


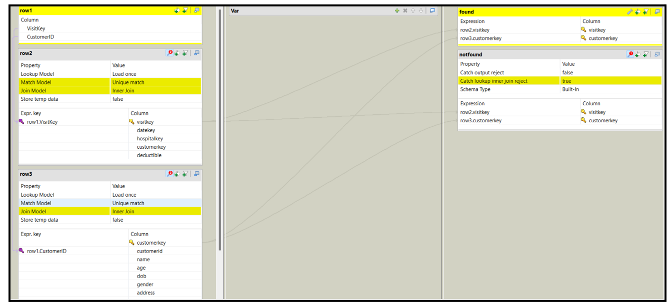
Cancer Symptom (bridge table)



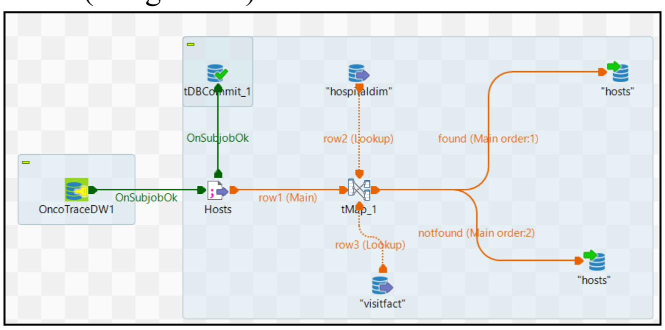


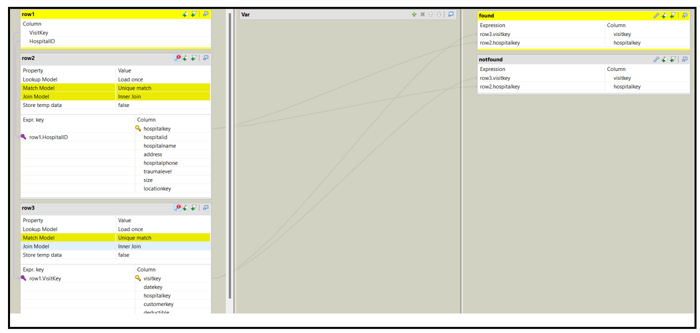
Partakes In (bridge table)



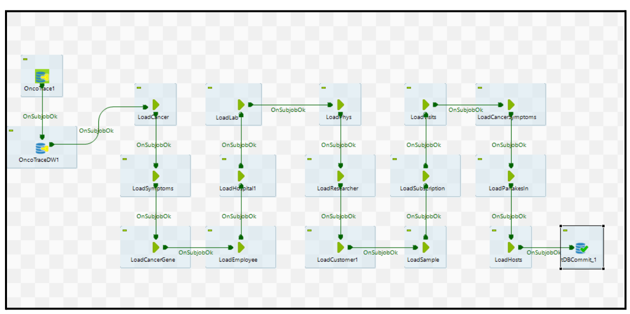


Hosts (bridge table)





Overall



Loaded Tables in PostgreSQL

Date Dimension

Query | Query History

```sql
select * from datedim
```

Data Output | Messages | Notifications

	datekey [PK] integer	date date	daynumber integer	weekday character varying (10)	weekdaynumber integer	monthnumber integer	monthname character varying (20)	year integer
1	1	2023-01-15	15	Sunday	7	1	January	2023
2	2	2023-02-10	10	Friday	5	2	February	2023
3	3	2023-03-05	5	Sunday	7	3	March	2023
4	4	2023-04-21	21	Friday	5	4	April	2023
5	5	2023-05-12	12	Friday	5	5	May	2023
6	6	2023-06-18	18	Sunday	7	6	June	2023
7	7	2023-07-07	7	Friday	5	7	July	2023
8	8	2023-08-13	13	Sunday	7	8	August	2023
9	9	2023-09-19	19	Tuesday	2	9	September	2023
10	10	2023-10-25	25	Wednesday	3	10	October	2023
11	11	2023-11-11	11	Saturday	6	11	November	2023
12	12	2023-12-03	3	Sunday	7	12	December	2023
13	13	2024-01-17	17	Wednesday	3	1	January	2024
14	14	2024-02-22	22	Thursday	4	2	February	2024
15	15	2024-03-29	29	Friday	5	3	March	2024
16	16	2024-04-04	4	Thursday	4	4	April	2024
17	17	2024-05-16	16	Thursday	4	5	May	2024
18	18	2024-06-30	30	Sunday	7	6	June	2024
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Location Dimension

Query | Query History

```sql
select * from locationdim
```

Data Output | Messages | Notifications

	locationkey [PK] integer	country character varying (50)	city character varying (50)
1	296	Russia	Tim
2	297	Russia	Zverevo
3	298	Indonesia	Sidomukti
4	299	Russia	Kolpashevo
5	300	Mexico	Primero de Mayo
6	301	Indonesia	Cibeureum
7	302	Uruguay	Rafael Perazza
8	303	Sweden	Saltsjöbaden
9	304	Guatemala	Chiquimulilla
10	305	Ukraine	Novomykolayivka
11	306	Russia	Lotoshino
12	307	Indonesia	Culeng
13	308	China	Yanjiang
14	309	Finland	Uusikaupunki
15	310	Finland	Vuokatti
16	311	China	Fengdian
17	312	Poland	Regimin
18	313	China	Xuefu
19	314	China	Laingu

Cancer

Query | Query History

```sql
select * from cancerdim
```

Data Output | Messages | Notifications

	cancerkey [PK] integer	cancerid integer	cancername character varying (255)	type character varying (100)	survivalrate numeric (5,2)	incidence numeric (5,2)	prevalence numeric (5,2)
1	7	1	Astrocytoma	Brain	34.00	3.20	8.70
2	8	2	Meningioma	Brain	90.00	7.40	25.00
3	9	3	Oligodendroglioma	Brain	67.00	0.95	3.50
4	10	4	Ductal carcinoma in situ (DCIS)	Breast	98.00	17.00	120.00
5	11	5	Invasive breast cancer	Breast	90.00	124.80	300.00
6	12	6	Invasive lobular breast cancer	Breast	87.00	12.40	50.00

Symptom

Query | Query History

```sql
select * from symptomdim
```

Data Output | Messages | Notifications

	symptomkey [PK] integer	symptomid integer	description text
1	1	1	Personality changes
2	2	2	Seizures
3	3	3	Headaches
4	4	4	Nausea
5	5	5	Vision changes
6	6	6	Hearing loss
7	7	7	Smell loss
8	8	8	Speech impairment
9	9	9	Arm or leg weakness
10	10	10	Bloody nipple discharge
11	11	11	Breast lump
12	12	12	Flattened nipple
13	13	13	Breast color change
14	14	14	Breast size or shape change
15	15	15	Breast skin peeling
16	16	16	Breast swelling

Cancer Gene Dimension

Query | Query History

```sql
select * from cancergenedim
```

Data Output | Messages | Notifications

	genekey [PK] integer	geneid integer	hugoname character varying (50)	sequence character varying (200)	location character varying (200)	mutation character varying (50)	cancerkey integer
1	493	1	TP53	ATGCGTACGTCAGGTTAC	Chromosome 17	R175H	7
2	494	2	EGFR	CCTGACCTGAGCGAGTCT	Chromosome 7	L858R	7
3	495	3	IDH1	ATCGTAGCTGCTGACGAG	Chromosome 2	R132H	7
4	496	4	NF2	GTCTGAGCCGCTGACGCA	Chromosome 22	E494K	8
5	497	5	PTEN	CGGTCGACGACGGTACGT	Chromosome 10	R130Q	8
6	498	6	SMARCB1	GTAGGTCGATGCTAGCCG	Chromosome 22	Q36X	8
7	499	7	OLIG2	ATGACGAGGACGTGCGCT	Chromosome 21	R171C	9
8	500	8	TP53	ATGCCTGCTGAGACGATG	Chromosome 17	R248W	9
9	501	9	BRCA1	CAGCAGAGCCTGAGCTGC	Chromosome 17	1418delA	10
10	502	10	BRCA2	GTTTTCAGGACGTCCGGT	Chromosome 13	D2720H	10
11	503	11	MAP3K1	AGCTAGTGTAGCTACGAG	Chromosome 5	S157C	11
12	504	12	ERBB2	CAGCAGTGACGATGACGT	Chromosome 17	S310F	11
13	505	13	MUC16	ATGCGTGATCTCGACGAC	Chromosome 19	R110K	12
14	506	14	TP53	CAGGTGCCACGCTACGGT	Chromosome 17	R273C	12
15	507	15	KMT2C	GAGCAGTAGTCGACGTCA	Chromosome 7	Q1101K	12
16	508	16	ALK	TGTAGAGACAGGAGGCTG	Chromosome 2	F1174L	10
17	509	17	AKT1	AAGCCGCCGACCTGTGCA	Chromosome 14	E17K	11
18	510	18	CCND1	AGCTGAGGCGTCGACGCT	Chromosome 11	G101V	11
19	511	19	FGFR2	CTAGTACGACGCGCCGTA	Chromosome 10	N549K	12

Employee Dimension

Query | Query History

```sql
select * from employeedim
```

Data Output | Messages | Notifications

	employeekey [PK] integer	employeeid integer	ssn character varying (11)	employeename character varying (255)	location character varying (100)	role character varying (100)	address character varying (255)	email character varying (100)
1	14646	101	123-45-6789	John Doe	New York	Physician	123 Main St, New York, NY	john.doe@example.con
2	14648	102	987-65-4321	Jane Smith	Los Angeles	Researcher	456 Elm St, Los Angeles, CA	jane.smith@example.cc
3	14651	103	111-22-3333	Michael Johnson	Chicago	Customer Service	789 Maple St, Chicago, IL	michael.j@example.cor
4	14655	104	444-55-6666	Emily Davis	Houston	IT	321 Oak St, Houston, TX	emily.d@example.com
5	14660	105	222-33-4444	David Brown	Phoenix	Physician	654 Pine St, Phoenix, AZ	david.b@example.com
6	14666	106	555-66-7777	Sarah Wilson	Philadelphia	Researcher	987 Cedar St, Philadelphia, PA	sarah.w@example.com
7	14673	107	888-99-0000	James Miller	San Antonio	Customer Service	135 Birch St, San Antonio, TX	james.m@example.con
8	14681	108	666-77-8888	Jessica Taylor	San Diego	IT	246 Walnut St, San Diego, CA	jessica.t@example.com
9	14690	109	333-44-5555	Daniel Anderson	Dallas	Physician	357 Spruce St, Dallas, TX	daniel.a@example.com
10	14700	110	777-88-9999	Laura Thomas	San Jose	Researcher	468 Cherry St, San Jose, CA	laura.t@example.com
11	14711	111	222-11-4444	Kevin White	Austin	Customer Service	579 Willow St, Austin, TX	kevin.w@example.com
12	14723	112	444-33-2222	Michelle Harris	Jacksonville	IT	680 Fir St, Jacksonville, FL	michelle.h@example.cc
13	14736	113	999-88-7777	George Martin	Columbus	Physician	791 Ash St, Columbus, OH	george.m@example.co
14	14750	114	555-22-8888	Amanda Lee	Charlotte	Researcher	902 Dogwood St, Charlotte, NC	amanda.l@example.co
15	14765	115	333-66-5555	Joshua Hall	San Francisco	Customer Service	234 Magnolia St, San Francisco, CA	joshua.h@example.con
16	14781	116	888-00-1111	Lisa King	Indianapolis	IT	345 Palm St, Indianapolis, IN	lisa.k@example.com
17	14798	117	777-99-2222	Andrew Wright	Seattle	Physician	456 Bay St, Seattle, WA	andrew.w@example.co
18	14816	118	222-44-3333	Jessica Scott	Denver	Researcher	567 Sea St, Denver, CO	jessica.s@example.con

phonenumber character varying (20)	hiredate date	firstname character varying (50)	lastname character varying (50)
555-278-2988	2020-01-15	John	Doe
555-279-2989	2019-03-22	Jane	Smith
555-280-2990	2021-07-12	Michael	Johnson
555-281-2991	2022-02-20	Emily	Davis
555-282-2992	2018-11-01	David	Brown
555-283-2993	2020-09-15	Sarah	Wilson
555-284-2994	2021-04-30	James	Miller
555-285-2995	2021-08-14	Jessica	Taylor
555-286-2996	2021-01-01	Daniel	Anderson
555-287-2997	2020-06-19	Laura	Thomas
555-288-2998	2019-12-10	Kevin	White
555-289-2999	2022-10-11	Michelle	Harris
555-290-3000	2021-09-25	George	Martin
555-291-3001	2023-01-05	Amanda	Lee
555-292-3002	2020-11-17	Joshua	Hall
555-293-3003	2018-05-23	Lisa	King
555-294-3004	2019-04-01	Andrew	Wright
555-295-3005	2022-07-30	Jessica	Scott

Hospital Dimension

Query | Query History

```
1 select * from hospitaldim
```

Data Output | Messages | Notifications

	ospitalkey [PK] integer	hospitalid integer	hospitalname character varying (255)	address character varying (255)	hospitalphone character varying (20)	traumalevel integer	size character varying (50)	locationkey integer
1	75	101112	Memorial Sloan Kettering Cancer Center	1275 York Ave New York NY 10065	347-798-9213	0	514	330
2	76	101113	Mayo Clinic	200 1st St SW Rochester MN 55905	507-284-2511	1	1265	331
3	77	101114	Dana-Farber/Brigham and Womens Cancer Ce...	5 Francis Street Boston MA 02115	877-332-4294	1	793	332
4	78	101115	City of Hope Comprehensive Cancer Center	1500 E Duarte Rd Duarte CA 91010	800-826-4673	0	234	333
5	74	101111	University of Texas MD Anderson Cancer Center	1515 Holcombe Blvd Houston TX 77030	877-632-6789	1	721	329

Lab Dimension

Query | Query History

```
1 select * from labdim
```

Data Output | Messages | Notifications

	labkey [PK] integer	labid integer	labname character varying (255)	principalinvestigator character varying (255)	labsize character varying (50)	labtype character varying (100)	accreditation character varying (100)	hospitalkey integer
1	14	201111	"Genomics Research Lab"	"Dr. Alice Jennings"	"1200"	"research"	"ISO 15189"	74
2	15	201112	"Advanced Production Lab"	"Dr. Michael Carter"	"800"	"production"	"CLIA"	75
3	16	201113	"Cardiac Medical Lab"	"Dr. Sarah Taylor"	"950"	"medical"	"CAP"	76
4	17	201114	"Clinical Diagnostic Lab"	"Dr. John Stewart"	"700"	"diagnostic"	"ISO 17025"	77
5	18	201115	"Neuro Research Facility"	"Dr. Emily Hayes"	"600"	"research"	"CLIA"	78
6	19	201116	"Bioengineering Lab"	"Dr. Robert Brooks"	"1400"	"production"	"ISO 15189"	74
7	20	201117	"Oncology Medical Lab"	"Dr. Karen Thompson"	"900"	"medical"	"CAP"	75
8	21	201118	"Pathology Diagnostic Center"	"Dr. David Lee"	"750"	"diagnostic"	"CLIA"	76
9	22	201119	"Immunology Research Lab"	"Dr. Sophia Martin"	"500"	"research"	"ISO 17025"	77
10	23	201120	"Pharmaceutical Production Lab"	"Dr. James Harris"	"1100"	"production"	"CLIA"	78
11	24	201121	"Molecular Biology Lab"	"Dr. Brian White"	"950"	"research"	"CAP"	74
12	25	201122	"Genetic Diagnostics Lab"	"Dr. Olivia Green"	"850"	"diagnostic"	"ISO 17025"	76

Physician Dimension

Query | Query History

```
1 select * from physiciandim
```

Data Output | Messages | Notifications

	physiciankey [PK] integer	physicianid integer	specialty character varying (100)	education character varying (255)	hospitalkey integer
1	10	14646	pediatric	University of Florida	74
2	11	14660	mastology	Ohio State University	75
3	12	14690	neruo	University of Exeter	76
4	13	14736	neruo	University of Siena	77
5	14	14798	pediatric	Johns Hopkins University	78
6	15	14876	cardio	Strathmore University	74
7	16	14970	cardio	Post University of Waterbury	78
8	17	15080	pediatric	Universidad Católica Boliviana, La Paz	76
9	18	15206	pediatric	Tamil Nadu Dr. Ambedkar Law University	77
10	19	15348	cardio	Texas Tech University Health Science Center	78
11	20	15506	internal	Universidad Tecnológica Oteima	74
12	21	15680	cardio	All India Institute of Medical Sciences	75
13	22	15870	pediatric	Edward Waters College	76

Researcher Dimension

Query   Query History

```
1  select * from researcherdim
```

Data Output   Messages   Notifications

	researcherkey [PK] integer	researcherid integer	specialty character varying (100)	education character varying (255)	labkey integer
1	3	14648	mastology	Universitat Internacional de Catalunya	14
2	4	14666	internal	China Medical University Shenyang	15
3	5	14700	pediatric	Rhodes College	16
4	6	14750	internal	Fachhochschule Niederrhein	17
5	7	14816	pediatric	Universidad Americana	18
6	8	14898	internal	Dilla University	19
7	9	14996	pediatric	Valdosta State University	20
8	10	15110	mastology	The College of New Jersey	21
9	11	15240	pediatric	Universitas Proklamasi 45	22
10	12	15386	internal	University of Swaziland	23
11	13	15548	neruo	The Queen's University Belfast	24
12	14	15726	neruo	Grodno State Agrarian University	25
13	15	15920	neruo	Kawasaki University of Medical Care	14

Customer Dimension

Query | Query History

```sql
select * from customerdim
```

Data Output | Messages | Notifications

	customerkey [PK] integer	customerid integer	name character varying (255)	age integer	dob character varying (20)	gender character varying (20)	address character varying (255)	email character varying (100)	phonenumber character varying (20)
1	1417	2934	Sammy Tott	74	1949-05-30	Male	83 Chinook Drive	stott0@diigo.com	830-793-9169
2	1418	6390	Enrique Kirsopp	33	1990-07-13	Bigender	14594 Ryan Avenue	ekirsopp1@devhub.com	415-381-7997
3	1419	9558	Garrik McIlhagga	84	1940-01-12	Male	921 Anniversary Street	gmcilhagga2@amazon.de	300-991-0541
4	1420	2578	Virgil Sorensen	65	1958-10-12	Male	76 Vidon Circle	vsorensen3@mac.com	319-139-0798
5	1421	7093	Charleen Vane	79	1944-02-07	Female	33 Tennessee Alley	cvane4@marriott.com	557-925-2254
6	1422	4318	Sherline Gorling	78	1945-07-26	Polygender	100 Blue Bill Park Drive	sgorling5@washington.edu	959-464-7522
7	1423	6297	Peirce Chomicz	49	1974-06-05	Male	62 Larry Place	pchomicz6@pen.io	491-544-4105
8	1424	1963	Stacee Gullick	77	1946-11-05	Male	40643 Onsgard Street	sgullick7@businesswire.com	761-582-6821
9	1425	3595	Lazar Woodwin	34	1989-01-26	Male	22743 Pawling Junction	lwoodwin8@newyorker.com	272-841-9240
10	1426	2593	Craggy Folkerd	47	1976-02-04	Male	471 Bellgrove Drive	cfolkerd9@opensource.org	171-104-6566
11	1427	4626	Delila Josefsson	48	1975-09-20	Female	169 Thackeray Alley	djosefssona@msu.edu	618-243-0560
12	1428	4241	Aldric Pic	27	1996-04-08	Male	67 Superior Hill	apicb@weibo.com	513-164-5848
13	1429	6748	Sholom Hounsom	79	1996-05-28	Male	1028 Fair Oaks Park	shounsomc@nature.com	792-817-9396
14	1430	1829	Carolynn Dingsdale	49	1996-03-16	Female	3 Eastlawn Parkway	cdingsdaled@parallels.com	712-280-0627
15	1431	648	Sax Roze	57	1966-11-27	Male	6 Thompson Pass	srozee@google.com	888-437-0444
16	1432	7028	Merrilee Irnis	66	1957-09-19	Agender	5500 Sheridan Plaza	mirnisf@feedburner.com	531-728-0970
17	1433	3764	Godfree Durston	55	1968-11-26	Male	0958 Springview Terrace	gdurstong@google.cn	663-247-1236
18	1434	7483	Jephthah Pitcock	29	1994-12-04	Male	30824 Rusk Lane	jpitcockh@canalblog.com	502-735-9602

Query | Query History

```sql
select * from customerdim
```

Data Output | Messages | Notifications

	dob character varying (20)	gender character varying (20)	address character varying (255)	email character varying (100)	phonenumber character varying (20)	previousdiagnosis text	bloodtype character varying (5)	smoker boolean	locationkey integer
1	1949-05-30	Male	83 Chinook Drive	stott0@diigo.com	830-793-9169	negative	A-	true	296
2	1990-07-13	Bigender	14594 Ryan Avenue	ekirsopp1@devhub.com	415-381-7997	positive	O-	false	297
3	1940-01-12	Male	921 Anniversary Street	gmcilhagga2@amazon.de	300-991-0541	negative	AB+	false	298
4	1958-10-12	Male	76 Vidon Circle	vsorensen3@mac.com	319-139-0798	positive	O+	false	299
5	1944-02-07	Female	33 Tennessee Alley	cvane4@marriott.com	557-925-2254	negative	O+	true	300
6	1945-07-26	Polygender	100 Blue Bill Park Drive	sgorling5@washington.edu	959-464-7522	negative	AB-	true	301
7	1974-06-05	Male	62 Larry Place	pchomicz6@pen.io	491-544-4105	negative	O+	true	302
8	1946-11-05	Male	40643 Onsgard Street	sgullick7@businesswire.com	761-582-6821	positive	A-	true	303
9	1989-01-26	Male	22743 Pawling Junction	lwoodwin8@newyorker.com	272-841-9240	positive	AB-	true	304
10	1976-02-04	Male	471 Bellgrove Drive	cfolkerd9@opensource.org	171-104-6566	at risk	O+	true	305
11	1975-09-20	Female	169 Thackeray Alley	djosefssona@msu.edu	618-243-0560	positive	B-	true	306
12	1996-04-08	Male	67 Superior Hill	apicb@weibo.com	513-164-5848	at risk	AB+	false	307
13	1996-05-28	Male	1028 Fair Oaks Park	shounsomc@nature.com	792-817-9396	positive	A-	false	308
14	1996-03-16	Female	3 Eastlawn Parkway	cdingsdaled@parallels.com	712-280-0627	negative	A-	true	309
15	1966-11-27	Male	6 Thompson Pass	srozee@google.com	888-437-0444	at risk	A-	true	310
16	1957-09-19	Agender	5500 Sheridan Plaza	mirnisf@feedburner.com	531-728-0970	positive	AB+	false	311
17	1968-11-26	Male	0958 Springview Terrace	gdurstong@google.cn	663-247-1236	negative	A-	true	312
18	1994-12-04	Male	30824 Rusk Lane	jpitcockh@canalblog.com	502-735-9602	at risk	O-	false	313

Sample Fact

Query | Query History

```sql
select * from samplefact
```

Data Output | Messages | Notifications

	genekey [PK] integer	datekey [PK] integer	customerkey [PK] integer	labkey [PK] integer	numberofgenemutations integer	predispositionlevel integer	resultstatus character varying (50)
1	493	2	1418	15	10	5	tested
2	494	6	1422	19	4	2	tested
3	495	10	1427	23	6	3	tested
4	496	13	1430	14	5	3	tested
5	497	17	1434	18	10	5	tested
6	498	22	1442	23	8	4	tested
7	499	26	1446	15	0	0	tested
8	500	28	1417	17	5	2	tested
9	501	33	1422	22	1	1	tested
10	502	37	1427	14	10	5	tested
11	503	41	1431	18	5	2	tested

Subscription Fact

Query | Query History

```
select * from subscriptionfact
```

Data Output | Messages | Notifications

	datekey [PK] integer	customerkey [PK] integer	subscriptiontype character varying (100)	status character varying (50)	fee integer	duration integer
1	52	1417	yearly	current	200	365
2	61	1418	monthly	ended	20	28
3	66	1419	yearly	renewed	200	366
4	48	1420	monthly	ended	20	29
5	68	1421	yearly	current	200	366
6	1	1422	yearly	renewed	200	365
7	57	1424	monthly	current	20	30
8	44	1425	yearly	ended	200	366
9	45	1426	monthly	current	20	31
10	46	1427	monthly	ended	20	31
11	67	1429	yearly	current	200	366
12	69	1430	yearly	renewed	200	366
13	54	1431	monthly	current	20	29
14	50	1432	yearly	current	200	366
15	53	1433	monthly	ended	20	30
16	65	1434	monthly	current	20	30
17	70	1436	yearly	renewed	200	365
18	60	1438	yearly	current	200	366
19	49	1440	monthly	ended	20	30

Visit Fact

Query | Query History

```
select * from visitfact
```

Data Output | Messages | Notifications

	visitkey [PK] integer	datekey integer	hospitalkey integer	customerkey integer	deductible integer
1	36	1	74	1417	100
2	37	2	75	1418	120
3	38	3	76	1419	300
4	39	109	74	1419	151
5	40	5	78	1421	212
6	41	6	74	1422	651
7	42	107	75	1422	321
8	43	122	76	1424	111
9	44	121	77	1425	512
10	45	123	78	1426	142
11	46	117	74	1426	101
12	47	103	75	1428	152
13	48	99	76	1429	120
14	49	111	78	1429	124
15	50	101	78	1431	245
16	51	100	74	1432	264
17	52	108	76	1434	124
18	53	89	78	1436	125
19	54	106	77	1436	154

Cancer Symptoms Bridge Table

Query Query History

```
select * from cancersymptomsdim
```

Data Output Messages Notifications

	symptomkey [PK] integer	cancerkey [PK] integer
1	1	7
2	2	7
3	3	7
4	4	7
5	2	8
6	3	8
7	4	8
8	5	8
9	6	8
10	7	8
11	8	8
12	9	8
13	2	9
14	3	9
15	9	9
16	11	10
17	10	10
18	11	11
19	12	11

Partakes In Bridge Table

Query Query History

Execute script F5

```
select * from partakesin
```

Data Output Messages Notifications

	visitkey integer	customerkey integer
1	36	1417
2	37	1418
3	38	1419
4	39	1419
5	40	1421
6	41	1422
7	42	1422
8	43	1424
9	44	1425
10	45	1426
11	46	1426
12	47	1428
13	48	1429
14	49	1429
15	50	1431
16	51	1432
17	52	1434
18	53	1436
19	54	1436

Hosts Bridge Table

Query | Query History

```
select * from hosts
```

Data Output | Messages | Notifications

	visitkey integer	hospitalkey integer
1	36	74
2	37	75
3	38	76
4	39	76
5	40	78
6	41	74
7	42	75
8	43	76
9	44	77
10	45	78
11	46	74
12	47	75
13	48	76
14	49	78
15	50	78
16	51	74
17	52	76
18	53	78
19	54	77

Incremental Load

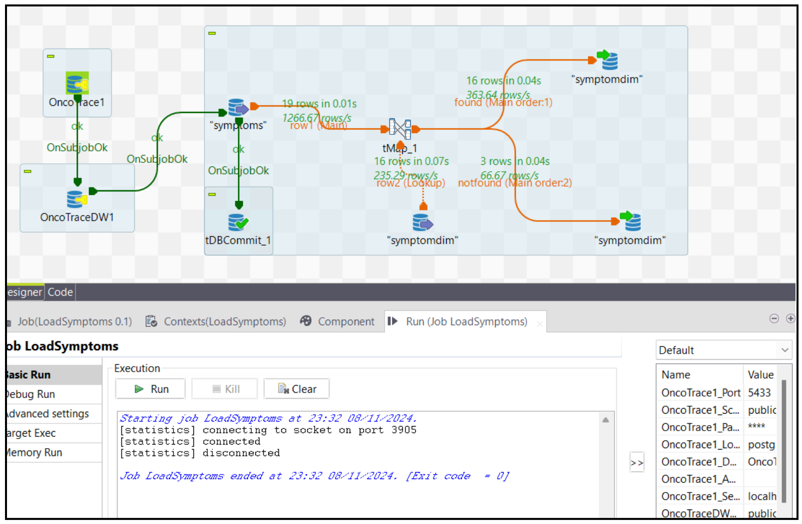
There were originally 16 rows in symptomdim. Three new rows were added, which can be seen being loaded in the notfound branch in the Talend job.

Query Query History

```
1 select * from symptomdim
```

Data Output Messages Notifications

	symptomkey [PK] integer	symptomid integer	description text
1	1	1	Personality changes
2	2	2	Seizures
3	3	3	Headaches
4	4	4	Nausea
5	5	5	Vision changes
6	6	6	Hearing loss
7	7	7	Smell loss
8	8	8	Speech impairment
9	9	9	Arm or leg weakness
10	10	10	Bloody nipple discharge
11	11	11	Breast lump
12	12	12	Flattened nipple
13	13	13	Breast color change
14	14	14	Breast size or shape change
15	15	15	Breast skin peeling
16	16	16	Breast swelling



After running the load job, the new rows are in the data warehouse:

Query Query History

```
1 select * from symptomdim
```

Data Output Messages Notifications

	symptomkey [PK] integer	symptomid integer	description text
1	1	1	Personality changes
2	2	2	Seizures
3	3	3	Headaches
4	4	4	Nausea
5	5	5	Vision changes
6	6	6	Hearing loss
7	7	7	Smell loss
8	8	8	Speech impairment
9	9	9	Arm or leg weakness
10	10	10	Bloody nipple discharge
11	11	11	Breast lump
12	12	12	Flattened nipple
13	13	13	Breast color change
14	14	14	Breast size or shape change
15	15	15	Breast skin peeling
16	16	16	Breast swelling
17	17	17	Fatigue
18	18	18	Pain
19	19	19	Bleeding

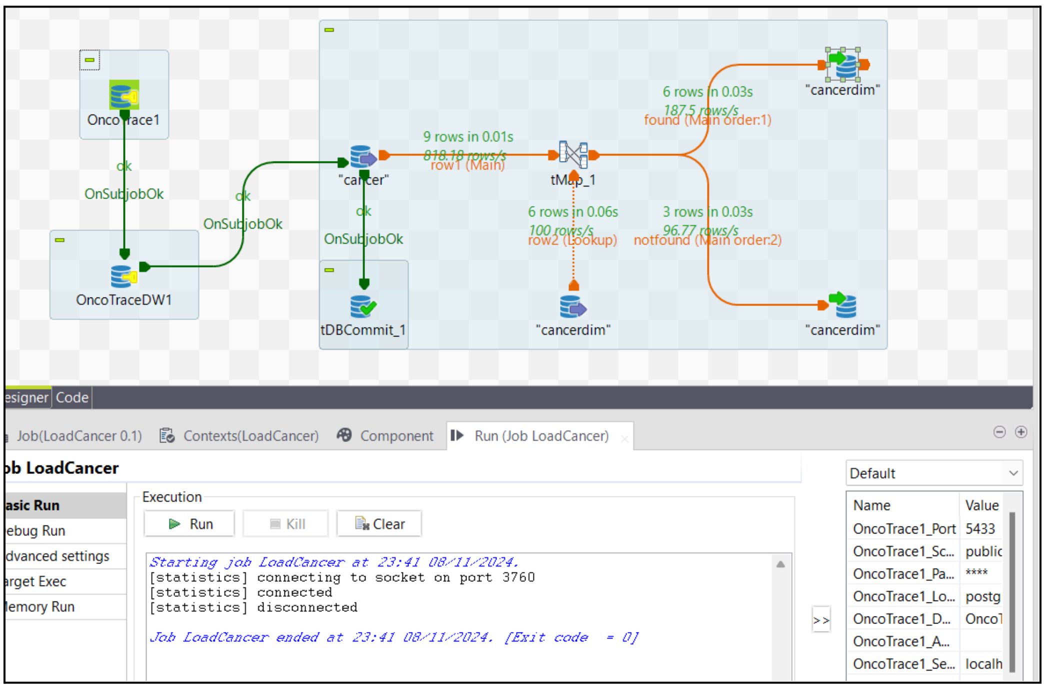
Here is another example with the cancerdim.

Query Query History

```
select * from cancerdim
```

Data Output Messages Notifications

	cancerkey [PK] integer	cancerid integer	cancername character varying (255)	type character varying (100)	survivalrate numeric (5,2)	incidence numeric (5,2)	prevalence numeric (5,2)
1	7	1	Astrocytoma	Brain	34.00	3.20	8.70
2	8	2	Meningioma	Brain	90.00	7.40	25.00
3	9	3	Oligodendroglioma	Brain	67.00	0.95	3.50
4	10	4	Ductal carcinoma in situ (DCIS)	Breast	98.00	17.00	120.00
5	11	5	Invasive breast cancer	Breast	90.00	124.80	300.00
6	12	6	Invasive lobular breast cancer	Breast	87.00	12.40	50.00



Query Query History

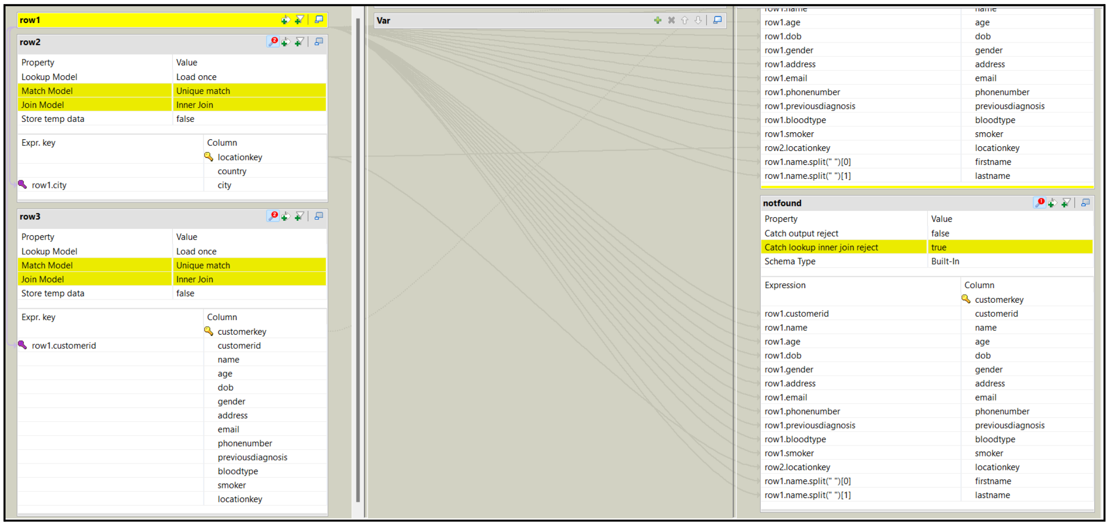
```
select * from cancerdim
```

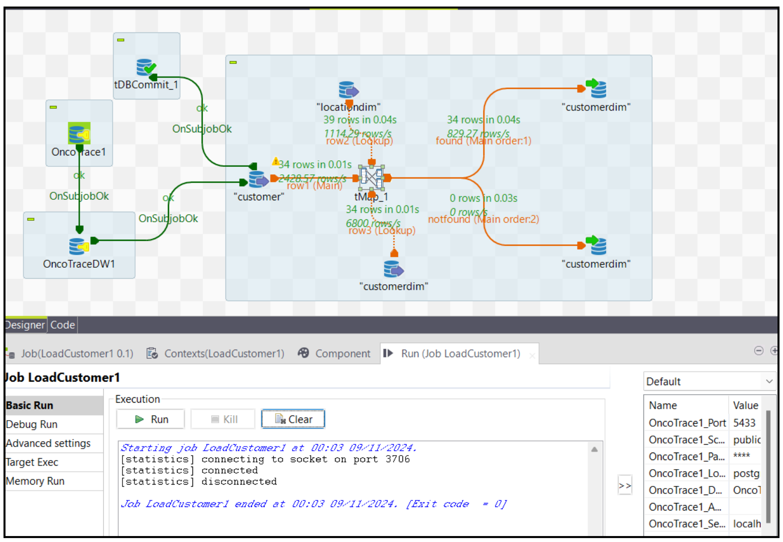
Data Output Messages Notifications

	cancerkey [PK] integer	cancerid integer	cancername character varying (255)	type character varying (100)	survivalrate numeric (5,2)	incidence numeric (5,2)	prevalence numeric (5,2)
1	7	1	Astrocytoma	Brain	34.00	3.20	8.70
2	8	2	Meningioma	Brain	90.00	7.40	25.00
3	9	3	Oligodendroglioma	Brain	67.00	0.95	3.50
4	10	4	Ductal carcinoma in situ (DCIS)	Breast	98.00	17.00	120.00
5	11	5	Invasive breast cancer	Breast	90.00	124.80	300.00
6	12	6	Invasive lobular breast cancer	Breast	87.00	12.40	50.00
7	13	7	Glioma	Brain	55.00	4.45	13.90
8	14	8	Medulloblastomas	Brain	45.00	3.67	10.11
9	15	9	Triple-Negative Breast Cancer	Breast	78.00	7.54	12.89

Transformations

Here we transform the Customer dimension by splitting ‘name’ into ‘firstname’ and ‘lastname.’





Query Query History

```
1 select * from customerdim
```

Data Output Messages Notifications

	customerkey [PK] integer	customerid integer	name character varying (255)	age integer	dob character varying (20)	gender character varying (20)	address character varying (255)	email character varying (100)	phonenumber character varying (2
1	1417	2934	Sammy Tott	74	1949-05-30	Male	83 Chinook Drive	stott0@diigo.com	830-793-9169
2	1418	6390	Enrique Kirsopp	33	1990-07-13	Bigender	14594 Ryan Avenue	ekirsopp1@devhub.com	415-381-7997
3	1419	9558	Garrik McIlhagga	84	1940-01-12	Male	921 Anniversary Street	gmcilhagga2@amazon.de	300-991-0541
4	1420	2578	Virgil Sorensen	65	1958-10-12	Male	76 Vidon Circle	vsorensen3@mac.com	319-139-0798
5	1421	7093	Charleen Vane	79	1944-02-07	Female	33 Tennessee Alley	cvane4@marriott.com	557-925-2254
6	1422	4318	Sherline Gorling	78	1945-07-26	Polygender	100 Blue Bill Park Drive	sgorling5@washington.edu	959-464-7522
7	1423	6297	Peirce Chomicz	49	1974-06-05	Male	62 Larry Place	pchomicz6@pen.io	491-544-4105
8	1424	1963	Stacee Gullick	77	1946-11-05	Male	40643 Onsgard Street	sgullick7@businesswire.com	761-582-6821
9	1425	3595	Lazar Woodwin	34	1989-01-26	Male	22743 Pawling Junction	lwoodwin8@newyorker.com	272-841-9240
10	1426	2593	Craggy Folkerd	47	1976-02-04	Male	471 Bellgrove Drive	cfolkerd9@opensource.org	171-104-6566
11	1427	4626	Delila Josefsson	48	1975-09-20	Female	169 Thackeray Alley	djosefssona@msu.edu	618-243-0560
12	1428	4241	Aldric Pic	27	1996-04-08	Male	67 Superior Hill	apicb@weibo.com	513-164-5848
13	1429	6748	Sholom Hounsom	79	1996-05-28	Male	1028 Fair Oaks Park	shounsomc@nature.com	792-817-9396
14	1430	1829	Carolynn Dingsdale	49	1996-03-16	Female	3 Eastlawn Parkway	cdingsdaled@parallels.com	712-280-0627
15	1431	648	Sax Roze	57	1966-11-27	Male	6 Thompson Pass	srozee@google.com	888-437-0444
16	1432	7028	Merrilee Irnis	66	1957-09-19	Agender	5500 Sheridan Plaza	mirnisf@feedburner.com	531-728-0970
17	1433	3764	Godfree Durston	55	1968-11-26	Male	0958 Springview Terrace	gdurstong@google.cn	663-247-1236
18	1434	7483	Jephthah Pitcock	29	1994-12-04	Male	30824 Rusk Lane	jpitcockh@canalblog.com	502-735-9602

Query Query History

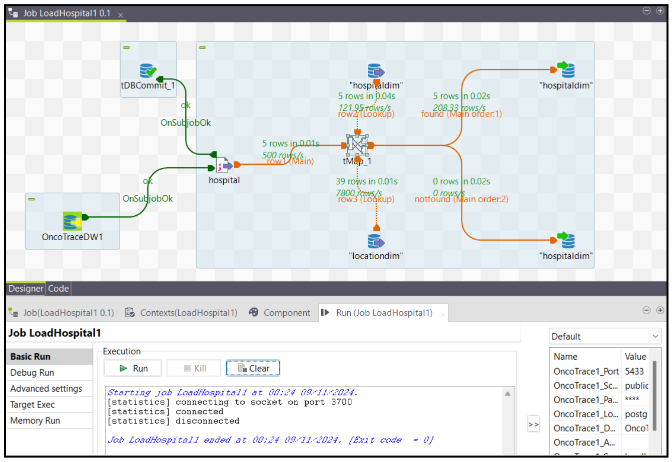
```
1 select * from customerdim
```

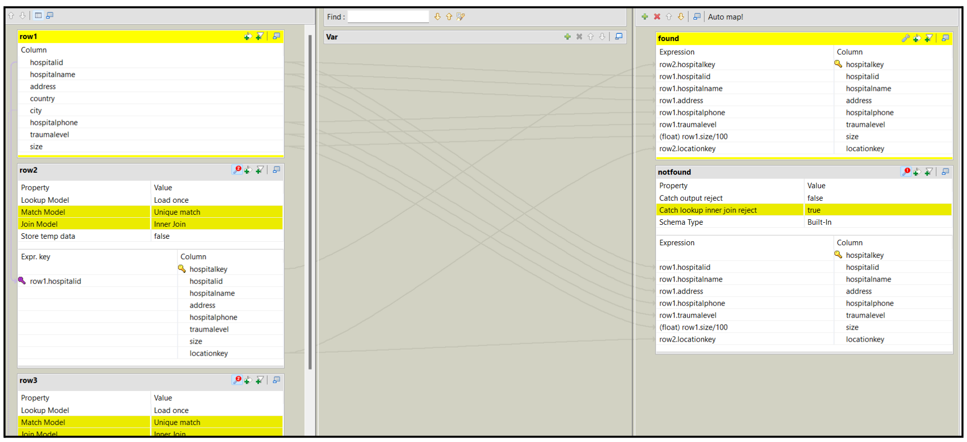
Data Output Messages Notifications

	varying (255)	email character varying (100)	phonenumber character varying (20)	previousdiagnosis text	bloodtype character varying (5)	smoker boolean	locationkey integer	firstname character varying (50)	lastname character varying (50)
1	ok Drive	stott0@diigo.com	830-793-9169	negative	A-	true	296	Sammy	Tott
2	an Avenue	ekirsopp1@devhub.com	415-381-7997	positive	O-	false	297	Enrique	Kirsopp
3	versary Street	gmcilhagga2@amazon.de	300-991-0541	negative	AB+	false	298	Garrik	McIlhagga
4	Circle	vsorensen3@mac.com	319-139-0798	positive	O+	false	299	Virgil	Sorensen
5	ssee Alley	cvane4@marriott.com	557-925-2254	negative	O+	true	300	Charleen	Vane
6	Bill Park Drive	sgorling5@washington.edu	959-464-7522	negative	AB-	true	301	Sherline	Gorling
7	Place	pchomicz6@pen.io	491-544-4105	negative	O+	true	302	Peirce	Chomicz
8	isgard Street	sgullick7@businesswire.com	761-582-6821	positive	A-	true	303	Stacee	Gullick
9	wling Junction	lwoodwin8@newyorker.com	272-841-9240	positive	AB-	true	304	Lazar	Woodwin
10	rove Drive	cfolkerd9@opensource.org	171-104-6566	at risk	O+	true	305	Craggy	Folkerd
11	keray Alley	djosefssona@msu.edu	618-243-0560	positive	B-	true	306	Delila	Josefsson
12	ior Hill	apicb@weibo.com	513-164-5848	at risk	AB+	false	307	Aldric	Pic
13	Oaks Park	shounsomc@nature.com	792-817-9396	positive	A-	false	308	Sholom	Hounsom
14	vn Parkway	cdingsdaled@parallels.com	712-280-0627	negative	A-	true	309	Carolynn	Dingsdale
15	son Pass	srozee@google.com	888-437-0444	at risk	A-	true	310	Sax	Roze
16	ridan Plaza	mirnisf@feedburner.com	531-728-0970	positive	AB+	false	311	Merrilee	Irnis
17	ingview Terrace	gdurstong@google.cn	663-247-1236	negative	A-	true	312	Godfree	Durston
18	isk Lane	jpitcockh@canalblog.com	502-735-9602	at risk	O-	false	313	Jephthah	Pitcock

Another example of a transformation can be seen in the initial load of the Employee dimension. Here first and last names are split. Refer to the Employee load on page 11 for the tMap configuration.

Here we transform the hospital dimension by expressing the 'size' column in hundreds (by dividing by 100).





Before transformation:

```
select * from hospitaldim
```

	ospitalkey [PK] integer	hospitalid integer	hospitalname character varying (255)	address character varying (255)	hospitalphone character varying (20)	traumalevel integer	size character varying (50)	locationkey integer
1	75	101112	Memorial Sloan Kettering Cancer Center	1275 York Ave New York NY 10065	347-798-9213	0	514	330
2	76	101113	Mayo Clinic	200 1st St SW Rochester MN 55905	507-284-2511	1	1265	331
3	77	101114	Dana-Farber/Brigham and Womens Cancer Ce...	5 Francis Street Boston MA 02115	877-332-4294	1	793	332
4	78	101115	City of Hope Comprehensive Cancer Center	1500 E Duarte Rd Duarte CA 91010	800-826-4673	0	234	333
5	74	101111	University of Texas MD Anderson Cancer Center	1515 Holcombe Blvd Houston TX 77030	877-632-6789	1	721	329

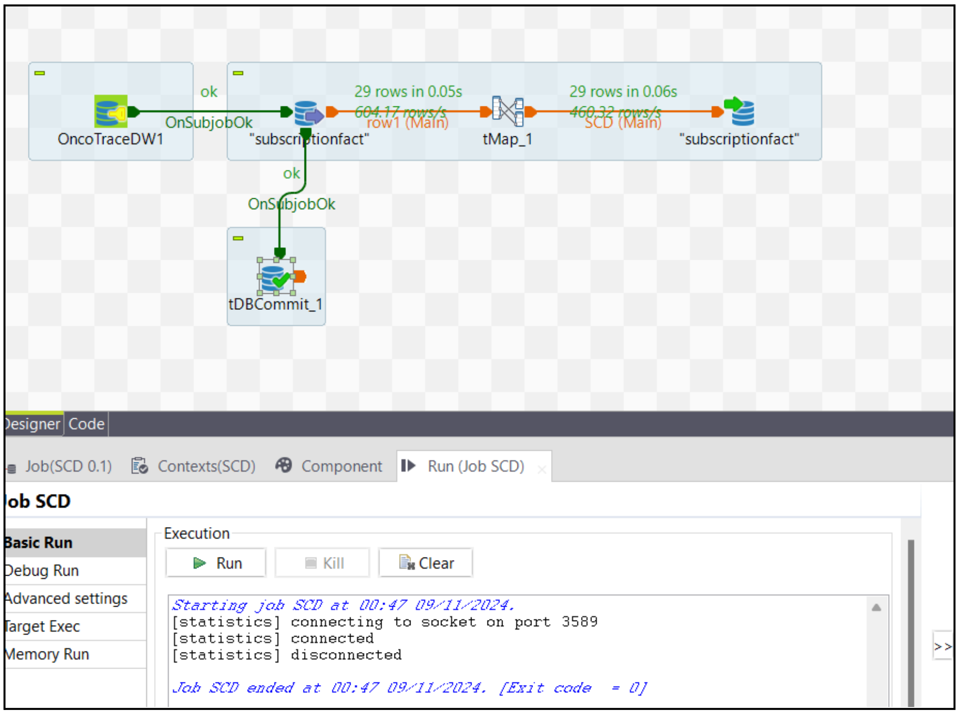
After transformation:

```
select * from hospitaldim
```

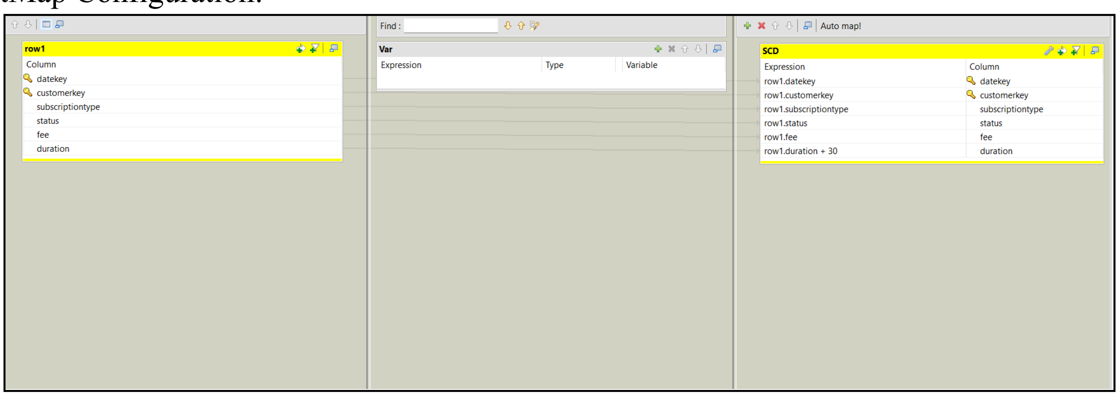
	hospitalkey [PK] integer	hospitalid integer	hospitalname character varying (255)	address character varying (255)	hospitalphone character varying (20)	traumalevel integer	size character varying (50)	locationkey integer
1	74	101111	University of Texas MD Anderson Cancer Center	1515 Holcombe Blvd Houston TX 77030	877-632-6789	1	7.21	329
2	75	101112	Memorial Sloan Kettering Cancer Center	1275 York Ave New York NY 10065	347-798-9213	0	5.14	330
3	76	101113	Mayo Clinic	200 1st St SW Rochester MN 55905	507-284-2511	1	12.65	331
4	77	101114	Dana-Farber/Brigham and Womens Cancer Ce...	5 Francis Street Boston MA 02115	877-332-4294	1	7.93	332
5	78	101115	City of Hope Comprehensive Cancer Center	1500 E Duarte Rd Duarte CA 91010	800-826-4673	0	2.34	333

SCD

Here we implement an SCD Type 1 when changes are made to ‘duration’ in the subscriptionfact table. We add 30 days after a month has passed, and the data is overwritten.



tMap Configuration:



Before SCD Type 1 updates:

Query | Query History

```
1  select * from subscriptionfact
```

Data Output | Messages | Notifications

	datekey [PK] integer	customerkey [PK] integer	subscriptiontype character varying (100)	status character varying (50)	fee integer	duration integer
1	52	1417	yearly	current	200	365
2	61	1418	monthly	ended	20	28
3	66	1419	yearly	renewed	200	366
4	48	1420	monthly	ended	20	29
5	68	1421	yearly	current	200	366
6	1	1422	yearly	renewed	200	365
7	57	1424	monthly	current	20	30
8	44	1425	yearly	ended	200	366
9	45	1426	monthly	current	20	31
10	46	1427	monthly	ended	20	31
11	67	1429	yearly	current	200	366
12	69	1430	yearly	renewed	200	366
13	54	1431	monthly	current	20	29
14	50	1432	yearly	current	200	366
15	53	1433	monthly	ended	20	30
16	65	1434	monthly	current	20	30
17	70	1436	yearly	renewed	200	365
18	60	1438	yearly	current	200	366
19	49	1440	monthly	ended	20	30

After SCD Type 1 updates:

Query | Query History

```
1  select * from subscriptionfact
```

Data Output | Messages | Notifications

	datekey [PK] integer	customerkey [PK] integer	subscriptiontype character varying (100)	status character varying (50)	fee integer	duration integer
1	52	1417	yearly	current	200	395
2	61	1418	monthly	ended	20	58
3	66	1419	yearly	renewed	200	396
4	48	1420	monthly	ended	20	59
5	68	1421	yearly	current	200	396
6	1	1422	yearly	renewed	200	395
7	57	1424	monthly	current	20	60
8	44	1425	yearly	ended	200	396
9	45	1426	monthly	current	20	61
10	46	1427	monthly	ended	20	61
11	67	1429	yearly	current	200	396
12	69	1430	yearly	renewed	200	396
13	54	1431	monthly	current	20	59
14	50	1432	yearly	current	200	396
15	53	1433	monthly	ended	20	60
16	65	1434	monthly	current	20	60
17	70	1436	yearly	renewed	200	395
18	60	1438	yearly	current	200	396
19	49	1440	monthly	ended	20	60

General Talend Documentation

Data Flow

Talend nodes are **bolded** while components are *italicized*. Tables are in SMALL CAPS. Columns are put in (parentheses). The OLAP CANCER table and snowflake CANCERDIM table are used as an example.

- Database connection **postgre_OLAP_connection** created to connect to the source, our PostgreSQL operational (OLAP) database, "Oncotrace."
  - The schema was retrieved and the entire public schema was selected.
- Database connection **postgre_snowflake_connection** created to connect to the target, our PostgreSQL relational (snowflake) database, "OncoTraceDW (snowflake)."
- For each table a job was created to load the OLAP table's data into the snowflake table.
  - **postgre_OLAP_connection** used as *tDBConnection(PostgreSQL)*.
  - **Postgre_snowflake_connection** used as *tDBConnection(PostgreSQL)*.
  - Data from CANCER used as *tDBInput(PostgreSQL)*.
  - Data from CANCERDIM used as *tDBOutput(PostgreSQL)*.
- Because CANCERDIM has a column (cancerkey) and CANCER doesn't, we generate it:
  - **tMap** component dragged onto the board.
  - **CANCER** connected to the **tMap** node.
  - Data from CANCERDIM used as *tDBInput(PostgreSQL)*.
    - This **CANCERDIM** connected to the **tMap**.
- In the **tMap** node, row1 (cancerid) was linked to row2 (cancerid).
  - row2 property settings set as follows:
    - Lookup Model: Load Once
    - Match Model: Unique match
    - Join Model: Inner join
    - Store temp data: false

Everything up to this point can be seen in Screenshot 1.

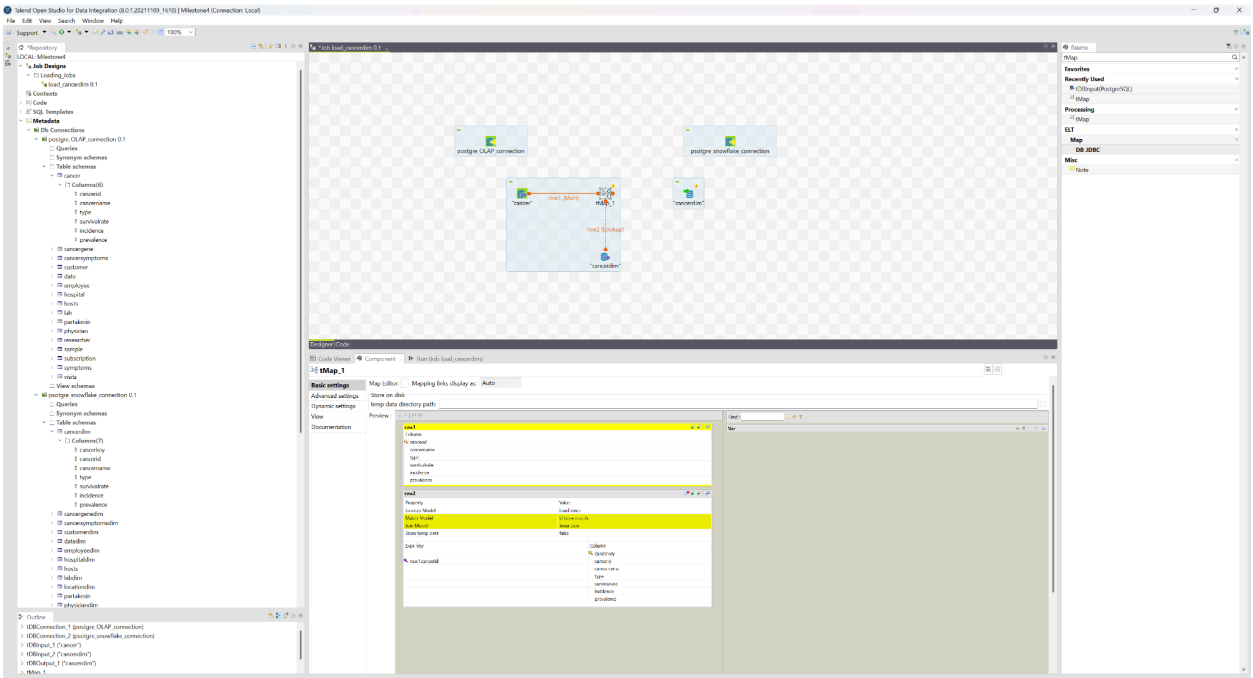
- Data from CANCERDIM used as another *tDBOutput(PostgreSQL)*.
  - **tMap** connected to the first **CANCERDIM** and named output "found"
  - **tMap** connected to the second **CANCERDIM** and named output "not_found"
- In the **tMap** node:
  - "found" property settings were set as follows:
    - Catch output reject: false
    - Catch lookup inner join reject: false
    - Schema Type: Built-In
  - "Not_found" property settings were set as follows:
    - Catch output reject: false
    - Catch lookup inner join reject: true
    - Schema Type: Built-In
  - Auto-Map! was used. This mapped row2 (cancerkey) to not_found (cancerkey). This is wrong, so this mapping was deleted, and for (cancerkey) in not_found, the default value was changed from nextval('cancerdim_cancerkey_seq'::regclass) to 0.

Control Flow

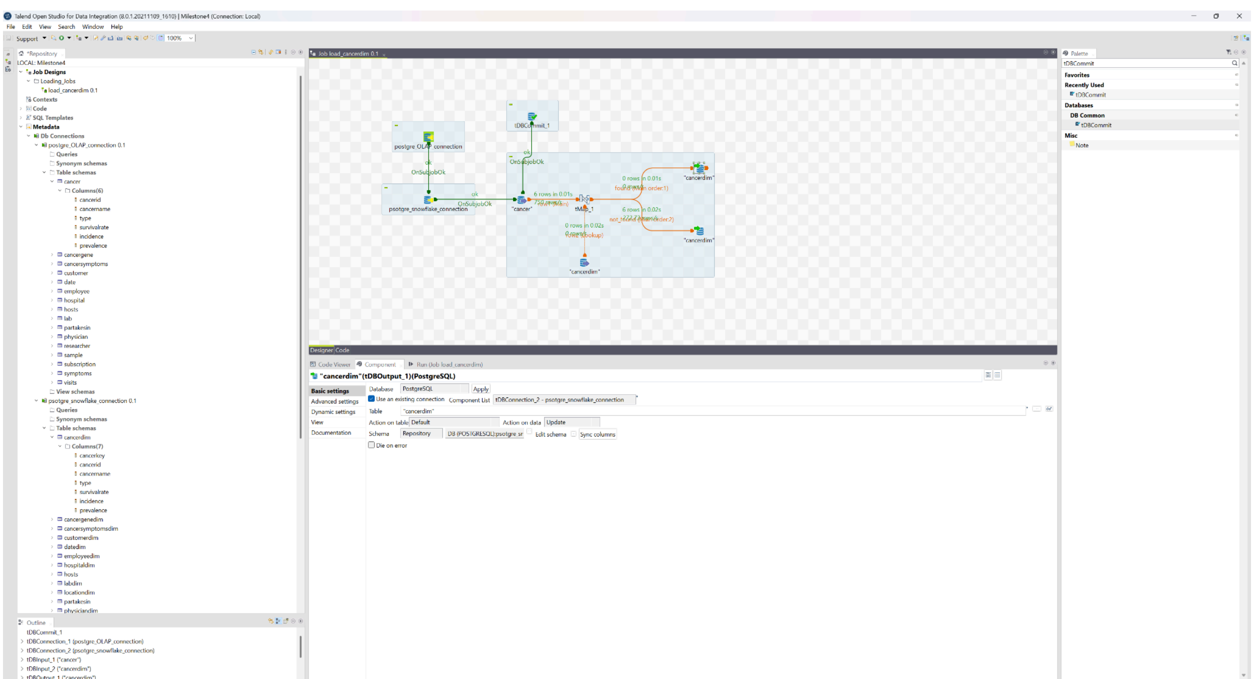
- **postgre_OLAP_connection** connected via "On Subjob OK" to **postgre_snowflake_connection**.
- **postgre_snowflake_connection** as connected via "On Subjob OK" to **CANCER**.
- Under **postgre_OLAP_connection** component basic settings, "Use or register a shared DB connection" selected, and the shared DB Connection was named: "postgre_OLAP_shared_connection"
    - Under **CANCER** component basic settings, "Use existing connection" selected, and the corresponding postgre_OLAP_connection was used.
- Under **postgre_snowflake_connection** component basic settings, "Use or register a shared DB connection" selected, and the shared DB Connection was named: "postgre_snowflake_shared_connection"
    - Under **CANCERDIM** component basic settings, "Use existing connection" selected, and the corresponding postgre_snowflake_connection was used.
- Under **CANCERDIM** "found" node basic settings, "Action on data" set to "Update."
- Under **CANCERDIM** "not_found" node advanced settings, a row named "cancerkey" was added to "Additional rows."
    - The "Position" set to "Replace."
    - The SQL expression set to "nextval('cancer_cancerkey_seq')"
- A **tDBCommit** component was dragged onto the board.
    - Under its component basic settings, the Database was selected to be Postgres
    - Component list was set to be our target, "postgre_snowflake_connection"
    - Close Connection was checked
- **CANCER** was connected to the tDBOutput node via an "On Subjob OK" Trigger

Everything up to here can be seen in Screenshot 2.

Talend Documentation Screenshots



Screenshot 1.



Screenshot 2.